

Whole-Body Robot Control: Paradigms, Learning Integration, and Future Frontiers

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Whole-body control coordinates high-degree-of-freedom robotic systems to achieve simultaneous dynamic stability and complex task execution. The field has transitioned from hierarchical, model-based optimization relying on simplified dynamics to unified, data-driven policies capable of adapting to unstructured environments. Despite these advances, a critical tension persists between the formal safety guarantees of analytical methods and the generalization capabilities of learning-based approaches. This survey synthesizes four distinct methodological paradigms: hierarchical optimization, end-to-end reinforcement learning, hybrid model-learning integration, and human-in-the-loop teleoperation frameworks. Analysis reveals that while pure learning offers superior adaptability, it lacks the rigorous constraint satisfaction required for safety-critical deployment, whereas traditional optimization struggles with model inaccuracies in dynamic scenarios. Consequently, emerging architectures increasingly fuse learned priors with model predictive control to balance computational tractability with robust performance. Future progress depends on establishing standardized benchmarks and developing hybrid systems that certify safety without sacrificing the flexibility needed for real-world autonomy.

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1 Introduction

The evolution of robotic autonomy has transformed whole-body control from a rigid engineering constraint into a dynamic challenge requiring the simultaneous regulation of high-degree-of-freedom systems [51, 189]. Early strategies relying on hierarchical decoupling increasingly fail to provide sufficient adaptability within unstructured environments, necessitating a transition toward unified frameworks that balance computational tractability with generalization capabilities. Contemporary research efforts now focus on synthesizing the precision of model-based optimization with the flexibility of data-driven learning to overcome the inherent brittleness of hand-crafted rewards and simplified template models [189]. This integration aims to resolve the tension between maintaining dynamic stability and executing complex tasks across diverse floating-base platforms.

The core methodologies can generally be categorized into several primary paradigms. 1) Hierarchical optimization frameworks dominate safety-critical applications by decomposing control into layered planning and tracking problems that regulate centroidal

momentum and schedule contacts via mixed-integer programming [51]. 2) Data-driven policies bypass explicit dynamics modeling to map observations directly to actions through reinforcement learning algorithms and adversarial motion priors [189]. 3) Hybrid integration strategies emerge as a critical middle ground, combining the robustness of analytical models with the adaptability of neural networks for residual correction or safety filtering [51, 189]. These distinct architectural choices define the current trade-off space between model fidelity, computational tractability, and behavioral flexibility.

An inherent adversarial tension defines the core problem, where the simultaneous pursuit of dynamic stability and complex task execution creates a scenario in which optimizing for one objective frequently destabilizes the other [51]. Model-based approaches often falter under the computational burden of high-dimensional optimization required for unstructured environments, while pure learning methods simultaneously suffer from a lack of formal safety guarantees during exploration. This conflict manifests as a continuous technological arms race, particularly in bridging the sim-to-real gap, where increased model

fidelity clashes directly with the necessity for rapid policy adaptation to physical hardware discrepancies.

Despite the theoretical promise of universal priors, current frameworks fail to achieve cross-embodiment generalization, forcing costly retraining for every new hardware morphology [189]. Existing behavior foundation models demonstrate promising zero-shot adaptation capabilities but remain critically constrained by severe data bottlenecks that limit training scale relative to language counterparts. Furthermore, the persistent sim-to-real dynamics discrepancy degrades policy performance during physical deployment, revealing a lack of robustness against sensor noise and actuator latency in learned controllers. Therefore, the primary motivation of this survey is to synthesize the evolution from hierarchical decoupled controllers to unified learning-based policies, identifying the critical trade-offs between computational tractability, dynamic performance, and generalization capabilities in unstructured environments.

Contributions. In this survey, we make three key contributions: **1) Unified Conceptual Framework:** We establish a cohesive definition of whole-body control that synthesizes the evolution from hierarchical decoupled controllers to unified learning-based policies, addressing the coordinated regulation of high-degree-of-freedom systems for dynamic stability and task execution as detailed in Sec. 2 and Sec. 3. **2) Critical Trade-off Analysis:** The work critically examines the inherent tensions between computational tractability, dynamic performance, and generalization capabilities across model-based optimization, data-driven learning, and human-in-the-loop integration frameworks discussed throughout Sec. 3. **3) Boundary Delineation and Gap Identification:** We explicitly define research boundaries spanning legged, humanoid, and mobile manipulator platforms while identifying critical gaps in theoretical guarantees for learning-based methods, thereby framing the challenges and future prospects outlined in Sec. 4.

Scope. To maintain analytical depth, this survey delineates its boundaries around four primary control paradigms: hierarchical optimization, end-to-end learning, hybrid model-learning integration, and human-in-the-loop teleoperation, as systematically categorized in Sec. 3. The physical embodiment scope is restricted to legged robots, humanoid platforms, and mobile manipulators that require dynamic whole-body coordination for stability and task execution. Conversely, high-level task planning, pure kinematic motion generation devoid of dynamics, and mechanical design optimizations fall strictly outside the defined boundaries of this work, ensuring a focused

examination of low-level control mechanisms rather than broader system architecture or abstract reasoning.

Survey Pipeline. The narrative begins by establishing the theoretical foundation, historical evolution, and evaluation standards necessary to contextualize the whole-body control problem in Sec. 2. Building upon this background, the core analysis in Sec. 3 categorizes existing methodologies into four distinct paradigms, dissecting their algorithmic mechanisms and performance bottlenecks. The discussion concludes in Sec. 4 by synthesizing critical deployment barriers and outlining a concrete research roadmap for future embodied autonomy. Finally, Sec. 5 summarizes the key insights derived from this comprehensive review.

2 Background

This chapter establishes the theoretical foundations necessary for analyzing whole-body control, beginning with a formal problem definition that frames dynamic locomotion and manipulation as a constrained optimization task involving floating-base dynamics Sec. 2.1. We then trace the historical evolution of control strategies, mapping the transition from simplified template models to modern frameworks utilizing full-order dynamics and learned priors Sec. 2.2. To ground these methodological discussions, we summarize current evaluation protocols, including motion capture datasets and simulation environments, while highlighting the scarcity of standardized benchmarks for loco-manipulation tasks Sec. 2.3. Finally, we delineate the boundaries between whole-body control and adjacent research domains such as motion planning and state estimation to clarify the interface where perception informs dynamic execution Sec. 2.4.

2.1 Problem Definition

Floating-Base Dynamics and State Space Formulation. Unifying floating-base kinematics with high-dimensional actuation topologies requires a state space formulation that spans rigid multi-legged structures to anatomically detailed musculoskeletal systems [27, 207]. The configuration manifold $\mathcal{Q}$ is parameterized by generalized coordinates $q \in \mathbb{R}^n$, which decompose into the floating base pose $x_b \in SE(3)$, joint angles θ_j , and, where applicable, muscle-tendon lengths l_{mt} . Governing dynamics extend the Euler-Lagrange formalism to incorporate discrete contact patterns and frictional constraints, linking centroidal

momentum evolution to internal shape changes via

$$M(q)\ddot{q} + C(q, \dot{q})\dot{q} + G(q) = S_u u + J_c^T \lambda_c \quad (1)$$

where $M(q)$ denotes the mass matrix, $C(q, \dot{q})$ captures Coriolis and centrifugal effects, $G(q)$ represents gravitational forces, S_u is the actuation selection matrix, and $J_c^T \lambda_c$ maps contact wrenches λ_c to joint space. To define the feasible trajectory space, this representation must accommodate heterogeneous embodiment specifics, such as lateral phase lags ϕ_{bl} in limbless-locomotor hybrids or hierarchical muscle synergy activations $a(t)$ in humanoids, thereby resolving kinematic redundancy through sensory feedback loops and duty factor β modulation.

Constrained Optimization Framework. Formulating whole-body control as a constrained optimization task requires minimizing task tracking errors and contact wrench deviations simultaneously within a quadratic programming structure [27, 35]. The objective function $J = \|\dot{x}_{task} - J\dot{q}\|^2 + \|\lambda_c - \lambda_{ref}\|^2$ balances kinematic fidelity against dynamic feasibility, where the Hessian matrix H and decision vector χ define the search space augmented by slack variables ϵ . Contact mechanics restrict this space through non-linear inequality constraints that confine ground reaction forces within friction cones approximated by $|\lambda_{tangential}| \leq \mu \lambda_{normal}$ while enforcing unilateral limits $\lambda_{normal} \geq 0$ to prevent penetration. The resulting feasible set constitutes a convex polytope in joint torque space, bounded by actuator saturation and dynamic equilibrium derived from centroidal momentum rates as formalized in

$$A_{ineq} \chi \leq b_{ineq}, \quad \tau_{min} \leq S_u^T (M\ddot{q} + h) - J_c^T \lambda_c \leq \tau_{max} \quad (2)$$

where Eq. (2) explicitly links the feasible torque region to the limits τ_{lim} and the rate of change of centroidal momentum $\dot{h}_G$, thereby defining the operational envelope for dynamic locomotion and manipulation.

Feasible Solution Space and Paradigm Mapping.

The admissible control manifold emerges from the intersection of dynamic equilibrium requirements and high-dimensional actuation bounds, where hierarchical decomposition strategies mitigate the computational complexity of resolving joint torques τ within limits τ_{lim} [35, 207]. Within this feasible set $\mathcal{F}$, distinct methodological pathways diverge: direct policy optimization operates in a latent action space $\mathcal{Z}$ using algorithms such as Soft Actor-Critic, whereas quadratic programming formulations explicitly enforce contact wrench feasibility inside friction cones $\mathcal{K}_\mu$. The coupling between centroidal momentum

rates $\dot{h}_G$ and internal shape changes further dictates the selection of stabilizers, ranging from model-based DCM-ZMP controllers for balance to data-driven priors that synthesize naturalistic motion under multimodal sensory uncertainty, thereby establishing the variable constraints that govern the optimization architectures detailed in Sections 3.1 and 3.2.

2.2 Technological History and Roadmap

From Template Models to Full-Order Dynamics.

Early locomotion controllers relied heavily on simplified template models, such as the Linear Inverted Pendulum, to enforce Zero-Moment Point stability constraints through heuristic gait generation and planar dynamics. This reliance on reduced-order approximations eventually gave way to frameworks incorporating full-order dynamics, where floating-base equations and contact wrench cones are embedded directly into non-linear optimization problems alongside centroidal momentum constraints. Contemporary developments have further transformed this field by replacing hand-crafted reward functions with foundation model priors, enabling control policies to generalize across diverse morphologies and tasks while facilitating sim-to-real transfer without extensive manual tuning.

Hybrid Stability and Intent Prediction.

Bridging the gap between rigid model-based stabilizers and flexible data-driven intent prediction enables operation in unstructured hazardous environments through generalized teleoperation architectures [35]. These frameworks employ cascaded retargeting and stabilization blocks that adapt reference trajectories using Divergent Component of Motion (DCM) and Zero-Moment Point (ZMP) criteria to preserve whole-body bilateral coupling while maintaining dynamic balance. To address latency-induced instability without compromising transparency, recent implementations integrate passivity-based wave variables or predictive displays directly into the communication channel.

Payload-Locomotion Trade-offs.

A fundamental tension exists between maximizing payload capacity and maintaining locomotion autonomy, particularly when shifting from dedicated manipulators to whole-body grasping strategies [49]. Historical analysis of static stability reveals that geometric constraints on object size relative to leg length and surface friction dictate feasibility across various contact configurations. While engaging the entire body enables handling objects exceeding robot dimensions, this capability traditionally necessitates halting dynamic

movement to satisfy these static requirements, effectively sacrificing continuous locomotion. Early model-based control frameworks struggled to resolve the resulting high-dimensional contact dynamics without compromising real-time feasibility, leaving a gap in adaptive manipulation over unstructured terrain.

2.3 Datasets, Metrics, and Benchmarks

From Capture to Synthesis. Reliance on expensive optical motion capture systems has given way to scalable synthetic datasets generated through optimal control principles, where Differential Dynamic Programming produces dynamically consistent trajectories and energy-efficient motion priors for multimodal training [14]. To maintain controller robustness against distribution shifts arising from composing independent skills, modern frameworks employ reinforcement learning with rollout-based data aggregation, utilizing domain randomization to ensure closed-loop skill execution within a shared whole-body policy [78]. Commercial gaming platforms now function as objective benchmarking environments, replacing traditional laboratory setups with reproducible metrics that correlate Just Dance scoring systems and kinematic retargeting accuracy against monocular SMPL pose reconstruction errors [76]. Scaling these efforts to massive parameter counts enables universal token spaces via vector quantization and masked prediction, unifying heterogeneous motion commands across embodiments for zero-shot generalization on training scales exceeding 100 million frames [107].

Bridging the Embodiment Gap. Wearable isomorphic exoskeletons address spatial scaling errors inherent in robot-free visual retargeting by capturing natural human kinematics and base velocities through direct joint-space alignment and LiDAR odometry [204]. Complementary robot-free interfaces employ online kinematic preview to ensure morphological feasibility during data collection, enabling hierarchical policy learning that maintains action chunk continuity without requiring physical hardware [118]. To further refine training distributions, privileged simulators filter large-scale motion capture datasets to remove dynamically infeasible trajectories, facilitating zero-shot sim-to-real transfer for whole-body tracking policies [56]. Heterogeneous co-training strategies subsequently extend semantic coverage by combining high-fidelity joint-level teleoperation with cheaper stationary data sources, though care must be taken to mitigate action MSE correlation failures when performing weight surgery on action layers [62].

Standardized Simulation Benchmarks. Moving beyond isolated locomotion tests, contemporary eval-

uation frameworks now integrate high-dimensional tactile sensing with dexterous manipulation to assess hierarchical control policies in unified environments. Recent suites employ platforms such as the Unitree H1 equipped with Shadow Hands within MuJoCo, utilizing 448 taxels and a 61-DoF action space to benchmark sample efficiency for algorithms like DreamerV3 and TD-MPC2 [148]. To facilitate cross-platform transfer, these benchmarks utilize canonical action interfaces, such as a 40D standard, which decouple high-level policy learning from low-level tracking dynamics. This architectural separation enables the evaluation of General Motion Trackers including SONIC and TWIST2 under perturbation-conditioned protocols that measure retention rates using diffusion policies and flow matching techniques conditioned on egocentric vision [169]. Furthermore, open-world assessment has evolved to include terrain-invariant sampling and two-stage curricula designed to test long-horizon reasoning on uneven ground. By fusing RGB and LiDAR data into semantic graphs, these frameworks allow vision-language models like GPT-4.1 and Qwen2.5-VL to execute geometry-constrained grasping and gait shaping through axis-aligned sampling strategies that bridge sim-to-real gaps [166].

Metric Paradigms. Evaluation protocols have transitioned from measuring purely kinematic tracking errors to employing physics-based stability metrics that filter dynamically infeasible reference motions prior to policy training [176]. This shift utilizes the distance between the Center of Mass and Center of Pressure alongside unstable sequence filtering within a bi-level optimization framework, often incorporating an adaptive tracking factor via exponential moving averages to ensure physical viability. Commercial game scoring systems now function as objective embodied AI benchmarks, revealing a distinct negative correlation between traditional position errors and holistic task success rates [76]. Specifically, Just Dance scores demonstrate a correlation coefficient ranging from negative 0.76 to negative 0.42 with PA-MPJPE in hardware-in-the-loop settings, highlighting a fundamental trade-off between stability and smoothness that practical deployment strategies must navigate. Advanced loco-manipulation assessment further requires decoupling kinematic redundancy exploration from dynamic execution to accurately measure end-effector precision amidst whole-body coordination conflicts [58]. By leveraging Kinematic Normalizing Flows for latent space navigation and partial reference tracking, researchers can isolate End-Effector Position Error from multimodal inverse kinematic distributions, providing a clearer metric for complex manipulation tasks.

2.4 Related Research Domains

Conceptual Boundary Between Planning and Execution. Distinct operational layers characterize the control stack, where high-level task planning manipulates discrete symbolic representations while dynamic execution governs continuous floating-base dynamics and contact constraints. This separation implies that abstract goals defined via hybrid automata must translate into feasible centroidal momentum trajectories without violating physical limits such as friction cones or wrench feasibility. Historical architectures often treated perception and control as decoupled pipelines, introducing latency that compromised reactive performance in unstructured environments. Modern approaches increasingly integrate state estimation directly into motion policies, enabling tight coupling where proprioceptive feedback and uncertainty propagation inform real-time adjustments to contact modes and trajectory parameterization.

Mechanical Architecture as a Control Constraint.

The dimensionality of the control problem is fundamentally dictated by whether manipulation degrees of freedom remain decoupled from or are embedded within the locomotion kinematic chain [49, 158]. Integrating actuators directly into leg structures reduces distal mass and rotational inertia I , yet this configuration imposes strict geometric limits on the reachable workspace and force-closure conditions defined by friction coefficient μ and object radius r . To compensate for reduced end-effector dexterity in such leg-mounted designs, whole-body coordination strategies must synchronize base trajectories with limb actuation, utilizing jerk-limited generation to maintain contact wrench feasibility during mode transitions. Theoretical analysis highlights a distinct trade-off where dedicated arms preserve locomotion autonomy at the cost of high inertial penalties, whereas whole-body grasping maximizes payload capacity but necessitates halting forward motion to satisfy static stability margins [49, 158].

Hierarchical Synergies and Multimodal Estimation.

Replicating biological fidelity in whole-body control necessitates decomposing high-dimensional actuation spaces into hierarchical synergies that mirror anatomical muscle groupings, moving beyond the independent regulation of individual degrees of freedom. Recent frameworks implement this through hierarchical actor-critic architectures where group-level actions modulate unit-level adjustments within muscle-tendon units, effectively managing systems with over 1,266 actuators via element-wise multiplication to capture spatiotemporal dynamics [207]. This structural decomposition couples tightly with mul-

timodal sensory integration, fusing binocular vision streams at 224×224 resolution, vestibular signals, and proprioceptive feedback to align simulated kinematics with experimental electromyography data. By employing stage-wise curriculum learning within Soft Actor-Critic optimizations, such approaches ensure that closed-loop state estimation remains consistent with physiological markers, bridging the gap between abstract dynamic execution and biological reality [207].

2.5 Discussion

Persistent reliance on high-fidelity simulation for training hierarchical policies induces a semantic-dynamic disconnect, wherein learned behaviors overfit to specific tracker dynamics rather than capturing generalizable physical intents [166, 169]. While analytical geometric frameworks offer rigorous stability guarantees for simplified planar models, they frequently fail to account for the three-dimensional inertial tipping modes essential to dynamic whole-body manipulation [27, 49]. Furthermore, the integration of ultra-high-dimensional musculoskeletal fidelity and teleoperation architectures introduces computational latencies and sensory simplifications that preclude real-time closed-loop adaptation in unstructured environments [35, 158, 207]. These compounding factors reveal a critical tension between model fidelity and computational speed, driving current research toward unified dynamics models that can bridge the gap between theoretical guarantees and heterogeneous embodiment requirements.

3 Methods

This section categorizes whole-body control methodologies into four distinct paradigms to analyze their algorithmic mechanisms and architectural choices. [Sec. 3.1](#) examines hierarchical optimization approaches that decompose control into layered problems for stability and constraint satisfaction. [Sec. 3.2](#) addresses end-to-end learning frameworks that derive policies directly from interaction data without explicit dynamics modeling. [Sec. 3.3](#) explores hybrid architectures integrating model-based robustness with learned adaptability through residual corrections or safety filters. Finally, [Sec. 3.4](#) discusses human-in-the-loop systems where teleoperation interfaces blend high-level intent with low-level robotic stability.

3.1 Hierarchical Optimization-Based Control

Hierarchical optimization-based control decomposes the complex problem of legged locomotion into manageable layers, typically separating high-level planning from low-level tracking to ensure stability and constraint satisfaction. This architectural approach relies on reduced-order representations to generate feasible momentum trajectories, as detailed in Sec. 3.1.1, while simultaneously managing discrete contact events through receding horizon strategies discussed in Sec. 3.1.2. To resolve conflicts between competing kinematic and dynamic objectives, lower control layers employ prioritized optimization schemes that enforce strict task hierarchies, a mechanism explored in Sec. 3.1.3. Furthermore, the integration of simplified template models provides computationally efficient references that capture essential dynamics without the burden of full-body complexity, as examined in Sec. 3.1.4. Together, these components form a cohesive framework that enables robots to execute dynamic maneuvers while adhering to physical limits.

3.1.1 Centroidal Dynamics & Momentum Regulation

Inertial Approximation via Reduced-Order Mass Distribution. Aggregating distributed limb masses into a minimal set of point entities enables analytical momentum regulation by approximating whole-body inertial properties through composite rigid body formulations [42, 43]. This abstraction facilitates the direct mapping of desired inertia tensors to joint configurations using closed-form geometric solvers, such as triangle-based limb approximations or dumbbell models, which bypass iterative numerical optimization to achieve microsecond-level latencies. Further efficiency gains arise from partitioning kinematic chains into constrained and unconstrained sets via centroidal projection, a strategy that decouples contact dynamics from free-limb motion within sequential lexicographic quadratic programming stages to yield a significant reduction in computation time [2].

Algebraic Complementarity and Wrench Consistency. Direct enforcement of complementarity constraints via differential dynamics replaces static relaxation to ensure exponential convergence during contact transitions without requiring predefined sequences [33]. This approach integrates Baumgarte stabilization with dynamic complementarity conditions, achieving high-fidelity rigid contact simulation through algebraic projection rather than horizon-based optimization. Unified centroidal momentum

optimization further couples these relaxed complementarity conditions with nonlinear actuation models, allowing the system to validate take-off feasibility and regularize cost functions while maintaining strict adherence to a given contact state [86]. By incorporating jet thrust integration, such formulations verify momentum consistency on platforms featuring non-trivial foot geometry and distributed mass [124]. These methods prioritize no-slip tasks and Center of Pressure constraints to minimize angular momentum, ensuring dynamic balance strictly through whole-body momentum regulation without relying on simplified template assumptions.

Integrable Coordinates for Momentum Decoupling. Decoupling global system regulation from non-integrable angular momentum dynamics is achieved by synthesizing integrable orientation coordinates through a least-squares approximation of the local connection matrix [21]. This approach utilizes a quaternion-based monomial basis to minimize the Frobenius norm, generating a history-independent coordinate that substantially reduces centroidal angular momentum oscillations while effectively damping z-axis instability. To further reduce execution latency, recent frameworks eliminate intermediate whole-body control layers by directly interfacing centroidal MPC policies with low-level joint impedance controllers [32]. Such architectures employ Sequential Linear Quadratic solvers to map feedforward torques via Recursive Newton-Euler Algorithm inverse dynamics, enabling high-frequency updates on complex humanoid systems. By focusing on these coordinate transformations and direct policy interfaces, the formulation isolates the geometric aspects of momentum regulation from lower-level constraint resolution mechanisms.

Inertial Coupling in Manipulation. Standard reduced-order planners often fail to maintain stability during dynamic manipulation when limb inertia constitutes a substantial fraction of total mass, necessitating explicit modeling of base-limb coupling to prevent unmodeled momentum exchange [151, 196]. This discrepancy becomes critical in systems where the limb mass ratio is significant, as center of mass fluctuations and underactuated direction exploitation degrade the fidelity of single rigid body dynamics approximations. To mitigate such instability, control strategies strictly enforce limb motion tracking over base references, ensuring that high-inertia limb movements do not compromise global balance despite the physical limitations of the centroidal model [196]. Complementary hardware co-design approaches address the root cause by minimizing the inertial gap through mass distribution optimization, such as con-

Table 1 Taxonomy of representative methods in **Hierarchical Optimization-Based Control**. This table compares hierarchical control methods by solver type and dynamics model, showing architectural diversity, and highlights the prevalence of centroidal approximations.

Method	Venue	Solver Type	Dynamics Model	Highlight
Centroidal Dynamics & Momentum Regulation (Sec. 3.1.1)				
Multi-Contact Control [151]	RA-L'21	SLQ-MPC	Augmented Centroidal	Augments centroidal dynamics with object states to capture base-limb-object inertial coupling during manipulation.
DCCs [33]	T-RO'22	Direct Multiple Shooting	Centroidal Kinematics	Enforces rigid contact via dynamic complementarity conditions, generating feasible momentum trajectories without predefined sequences.
Whole-Body MPC [32]	Humanoids'22	SLQ	Centroidal Dynamics	Directly maps centroidal MPC policies to joint impedance, minimizing execution discrepancy for high-DoF systems.
Paredes et al. [124]	ACC'24	Hierarchical QP	ALIP Centroidal	Regulates centroidal angular momentum hierarchically to enforce ALIP assumptions despite non-zero reaction moments.
Ahn et al. [2]	Internatio'24	Lexicographic QP	Reduced Centroidal	Projects unconstrained kinematic chains into centroidal space, decoupling contact dynamics from limb control in WBC.
Model Predictive Control with Contact Planning (Sec. 3.1.2)				
Unified Multi-Contact OC [151]	RA-L'21	SLQ-MPC	Augmented Centroidal	Augments centroidal dynamics with object states to unify locomotion and manipulation contact scheduling via switched-system SLQ.
Hybrid iLQR (HiLQR) MPC [80]	T-RO'22	Event-Driven iLQR	Full-Order Rigid	Propagates gradients across hybrid transitions using saltation matrices within event-driven iLQR for dynamic contact sequence optimization.
CAFE-MPC + VWBC [91]	IEEE'24	MS-iLQR	Multi-Fidelity Hybrid	Partitions horizon into high-fidelity leading and low-fidelity trailing plans, extracting value functions for constraint-satisfying whole-body control.
Contact-Explicit TAMP [28]	Humanoids'	Logic-Geometric Programming	Centroidal Dynamics	Unifies locomotion and manipulation via symbolic contact modes, jointly optimizing discrete sequences and continuous trajectories in TAMP.
Chen et al. [22]	arXiv'25	FDDP	Unified Multibody	Integrates point and line contacts into unified FDDP framework, eliminating discrete mode switching through geometry-aware kinematic mapping.
Quadratic Programming for Task Prioritization (Sec. 3.1.3)				
RHP-HQP [53]	IROS'22	Recursive HQP	Kinematic Constraints	Recursive Hierarchical Projection matrices enable smooth priority transitions without increasing quadratic programming solve counts.
SEIKO [143]	T-Mech'22	Single-Iteration SQP	Quasi-Static Equilibrium	Single-iteration Sequential Quadratic Programming with exponential penalty modulation achieves kilohertz contact switching rates.
Reduced-Accuracy QP [18]	Humanoids'22	Dual Active-Set QP	Full Rigid Body	Decoupling matrix update frequencies from control loops exploits solution robustness against injected numerical noise.
Ahn et al. [2]	Internatio'22	Lexicographic QP	Reduced Centroidal	Centroidal dynamic projection reduces dimensionality for sequential Lexicographic Quadratic Programming stages preserving task hierarchy.
Whole-Body Constrained [163]	RA-L'25	HQP	Full Rigid Body	Hierarchical Quadratic Programming integrates adjustable soft constraints for foot-terrain interaction alongside hard actuator limits.
Reduced-Order Template Models (Sec. 3.1.4)				
VI-MPC [7]	CDC'24	Convex MPC	Variable Inertia SRB	Integrates neural-network-predicted variable inertia into convex MPC to augment Single Rigid Body model fidelity.
Paredes et al. [124]	ACC'24	Hierarchical QP	ALIP Centroidal	Enforces ALIP assumptions on flat-footed robots via hierarchical momentum regulation and explicit contact wrench constraints.
PS-LIPM [150]	arXiv'25	MPC-QP	Hybrid DCM	Augments LIPM with piecewise slope dynamics and angular momentum coefficients for uneven terrain traversal.

centrating actuator mass near the system centroid or employing four-bar linkage transmissions to enhance centroidal inertia isotropy [121].

3.1.2 Model Predictive Control with Contact Planning

Combinatorial Complexity in Contact Sequence Optimization. Expanding the feasibility region for dynamic maneuvers requires treating contact switching instants as decision variables within a fixed mode sequence rather than adhering to rigid timing pre-

scriptions [70]. Gradient-based refinement across these hybrid transitions depends on accurate linearization of discontinuous dynamics, where saltation matrices and reset map Jacobians facilitate convergence during impact events [80]. To resolve acyclic locomanipulation tasks without predefined gait cycles, unified frameworks couple discrete symbolic graph search with continuous whole-body dynamics [28]. Alternatively, relaxing complementarity conditions in centroidal formulations allows contact sequences and aerial phases to arise directly from a single optimal control problem [86]. Addressing environmental un-

certainty, stochastic predictive control replaces heuristic constraint shrinking with covariance-propagated safety margins derived from chance constraints [47].

Computational Mitigation Strategies. To address the prohibitive cost of nonlinear long-horizon prediction, cascaded fidelity scheduling partitions the horizon into high-fidelity leading segments and low-fidelity tailing regions, utilizing coarse time steps or single-rigid-body approximations to estimate terminal costs without excessive overhead [91]. Structural consistency across contact transitions is further maintained by replacing discontinuous wrench formulations with Zero-Moment Point variables, a reformulation that enables machine learning-based warm-start strategies to achieve sub-millisecond latency within differential dynamic programming frameworks [77]. Solver efficiency is also tuned by strategically truncating iterative steps, such as limiting ADMM iterations or executing single-step SQP, while initial-timestep Control Barrier Functions enforce strict safety constraints despite the reduced computational effort [45, 75]. Finally, hierarchical planning operates within a reduced command space stabilized by pre-trained low-level policies, effectively mitigating the curse of dimensionality inherent in direct joint-space optimization for contact-rich maneuvers [193].

Robust Optimization for Model Mismatch. To address model mismatch within the prediction window without relying on external estimators, robust MPC formulations employ tube-based strategies that confine state trajectories within invariant sets derived from bounded disturbance assumptions [87, 88]. These approaches optimize nominal contact sequences while tightening constraints to guarantee feasibility across all possible realizations of parametric uncertainty. Complementing deterministic bounds, scenario-based methods sample multiple disturbance realizations to construct probabilistic safety margins, enabling the planner to balance conservatism against performance during dynamic contact transitions [116]. By embedding uncertainty directly into the optimal control problem, these techniques maintain valid contact modes and wrench distributions despite unmodeled dynamics or environmental variations.

Architectural Trade-offs: Contact-Implicit versus Contact-Explicit Formulations. Replacing discrete mode scheduling with continuous complementarity constraints allows contact-implicit formulations to generate contact sequences automatically, though this approach often introduces numerical stiffness and sensitivity to local minima [33]. Such methods typically rely on dynamic complementarity conditions approximated via techniques like Baumgarte stabilization or

hyperbolic secant functions, resulting in non-convex optimization problems that require careful initialization. Conversely, contact-explicit approaches encode switching logic directly into the problem structure or utilize unified kinematic mappings to eliminate discontinuous mode transitions entirely [22, 151]. These frameworks treat locomotion as a switched-system optimal control problem, enforcing equality constraints on specific contact modes or employing geometry-aware mappings within sequential linear-quadratic MPC schemes to maintain computational tractability.

3.1.3 Quadratic Programming for Task Prioritization

Null-Space Projection Mechanisms for Strict Hierarchical Enforcement. Recursive hierarchical projection matrices facilitate continuous task priority transitions by strictly preserving higher-priority components within the optimization subspaces of lower-level objectives [53]. Within Stack-of-Tasks frameworks, this architecture projects secondary kinematic constraints, including singularity avoidance and joint limits, into the null space of primary end-effector trajectories to guarantee uninterrupted execution [155]. Differential geometry-based policy composition extends this principle by utilizing prioritized Jacobians equipped with dynamically consistent pseudo-inverses, thereby enforcing strict behavior hierarchies without relying on iterative constrained optimization [1]. Furthermore, geometric projected gradient descent solvers maintain orthogonality between base stabilization wrenches and limb tracking objectives by explicitly projecting joint torques into the null space of the base mapping Jacobian [73].

Slack Variable Regularization and Constraint Satisfaction Trade-offs. Removing slack variables from hierarchical formulations enforces strict task precedence and reduces computational latency, though this approach risks discarding low-priority objectives when constraints become active [177]. Distinct constraint classifications address such rigidity by maintaining hard physical limits as binding during training while permitting dynamic adjustment of soft interaction constraints at deployment without policy retraining [163]. Furthermore, converting nonlinear performance indices into linear polyhedral inequalities via offline discretization enables their integration as acceleration-level soft constraints within online quadratic programs [165].

Computational Latency Management. Exploiting the inherent robustness of quadratic solutions allows the update frequency of Hessian and constraint ma-

trices to be decoupled from the control sampling rate, significantly lowering computational load without degrading tracking accuracy [18]. Dimensionality reduction further accelerates solve times by restricting optimization to active constraint subsets, which preserves essential interaction dynamics while reducing total degrees of freedom [2]. Direct reduction of the decision variable space is also achievable through analytical elimination of equality constraints via QR factorization prior to solver invocation [37]. Additionally, decomposing equations of motion to analytically eliminate joint torques from the decision vector enables single-iteration convergence for quasi-static equilibrium problems [143].

Oscillation Suppression and Stability Near Constraint Boundaries. Predictive impulse propagation constraints pre-emptively bound joint accelerations to suppress oscillations arising from abrupt state jumps during physical contact [172]. This approach utilizes worst-case impulse predictions derived from the coefficient of restitution and inverse operational space inertia matrix to enforce linear inequality constraints on joint accelerations, thereby limiting ZMP deviations. Complementing this feedforward protection, Center of Mass admittance tasks dynamically shift equilibrium references based on force errors to dampen instability near friction cone boundaries without requiring explicit trajectory replanning [142]. By integrating support function metrics with linearized friction cones, such controllers enable gravity-assisted force generation and online balance region updates driven by real-time feedback. Furthermore, encoding task hierarchies as quadratic inequality constraints within a receding horizon prevents priority inversion and ensures smooth convergence near active constraint limits [84]. Formulating these relationships as a convex QCQP allows for finite-horizon optimization using linearized dynamics around nominal trajectories, which can be solved recursively to maintain strict adherence to dynamic feasibility.

3.1.4 Reduced-Order Template Models

Computational Efficiency vs. Dynamic Fidelity. Modular frameworks decouple simplified template planners from whole-body controllers to enable rapid algorithmic swapping while maintaining standardized interface protocols [34]. These architectures utilize a unified state-space formulation supporting variants such as ALIP, H-LIP, MLIP, and DCM, interfacing with QP, ID, or IK solvers through hardware abstraction layers to minimize velocity tracking errors. Embedding offline stable gait trajectories as terminal costs allows nonlinear model predictive control to utilize significantly shorter prediction horizons

without sacrificing stability guarantees [45]. By leveraging Hybrid Zero Dynamics references within a sequential quadratic programming scheme solved via direct multiple shooting, this approach achieves horizon reductions to approximately 0.2s while preserving disturbance rejection capabilities. Extending template models with specific inertial parameters further captures underactuated coupling effects, enabling tractable trajectory generation via differential-flatness-like properties [19]. The W-SLIP model, which incorporates non-trivial wheel mass, facilitates polynomial coefficient optimization across stance and flight phases to expand region of attraction coverage through dedicated quadratic programming formulations.

Augmentation Strategies for Non-Linear Coupling and Multi-Body Dynamics. Explicit flywheel dynamics extend standard template formulations by capturing the angular momentum contribution of redundant limbs, thereby enabling stable execution of highly dynamic maneuvers without inflating model dimensionality [170]. To address inertial shifts arising from limb reconfiguration, recent architectures integrate configuration-dependent variable inertia predictors directly into convex optimization frameworks, correcting single rigid body approximations during high-speed turning [7]. Alternative approaches abstract full articulated dynamics into deformable centroid models, which preserve exact inertial coupling effects while maintaining the tractability required for low-dimensional trajectory optimization [15]. Finally, enforcing physical contact wrench constraints within hierarchical task controllers ensures that reduced-order planners remain valid for robots utilizing non-point contact feet and distributed mass distributions [124].

Validity Regions and Constraint Enforcement Mechanisms. Extending the operational envelope of linear templates requires hybrid divergence dynamics that explicitly model continuous center of mass height variations across piecewise gradients, enabling simultaneous optimization of step position and duration on non-periodic terrain [150]. To certify stability without precise Jacobian inversion, Lyapunov-based energy shaping exploits Hamiltonian structure to bound simulation errors between reduced-order models and full rigid-body dynamics, ensuring passivity against external disturbances even near singular configurations [82]. When ground reaction forces alone cannot satisfy physical feasibility limits on steep inclines, integrating auxiliary thrust vectors into quadratic programming formulations enforces friction cone constraints while maintaining computational efficiency for variable length inverted pendulum projections

[127].

Failure Modes from Contact Constraint Violations. Tracking divergence arises when simplified inertia mappings in analytical solvers encounter trunk orientation asymmetries that violate the assumed decoupling between yaw alignment and limb compensation [42]. Such failures stem from inertia product errors within five-mass models, where dumbbell tilting approximations and strict anthropomorphic structure assumptions break down under unique mass-to-joint correspondence conditions. Furthermore, hierarchically decoupling contact timing from full-body feasibility verification induces actuator saturation during high-impact landing phases because simplified rigid body dynamics neglect leg inertia effects [119]. In these scenarios, SRBD optimization relying on rotation matrix parameterization yields inaccurate torque predictions, particularly limiting QP landing controllers during partial foot contact events.

3.1.5 Discussion

Despite their unparalleled safety dominance in contact-rich scenarios derived from strict dynamic constraint enforcement, optimization-based frameworks face severe scalability limits when applied to high-degree-of-freedom systems [2, 10, 127]. The computational latency inherent in solving sequential linear quadratic programs often pushes real-time feasibility thresholds, where even minor sensor drift or torque control failures can destabilize the planner if computation times exceed critical bounds such as 0.3 ms. This rigidity is compounded by a fundamental adaptability gap arising from reliance on accurate reduced-order models; unmodeled inertial variations or complex loco-manipulation couplings significantly degrade tracking performance during agile maneuvers [7, 30, 170]. Discrepancies between variable centroidal inertia and simplified approximations like F-SLIP or H-LIPM deadbeat control disrupt step-to-step consistency, particularly when swing arm dynamics introduce angular momentum regulation challenges not captured by the template. Furthermore, tuning cost functions and task hierarchies remains a non-trivial, heuristic-dependent bottleneck that frequently necessitates manual reconfiguration for distinct environments or uncertainty levels [111, 133]. Adjusting adaptive MPC weighting matrices or hierarchical prioritization schemes often requires expert intuition to balance friction cone constraints against actuation limits, as automated methods struggle to generalize across diverse operational contexts without extensive recalibration.

3.2 End-to-End Learning-Based Control

End-to-end learning-based control bypasses explicit dynamics modeling by deriving policies directly from interaction data or expert demonstrations. This paradigm encompasses a spectrum of methodologies, ranging from reinforcement learning algorithms that optimize reward functions for complex motor skills to imitation learning frameworks that regularize behavior through adversarial motion priors. [Sec. 3.2.1](#) examines proximal policy optimization techniques tailored for coordinated loco-manipulation tasks, while [Sec. 3.2.2](#) analyzes how adversarial discriminators enforce physical feasibility and naturalness in learned motions. Recent advancements further integrate large-scale foundation models to bridge semantic understanding with low-level actuation, as detailed in [Sec. 3.2.3](#), alongside hierarchical approaches that decompose long-horizon objectives into reusable skill primitives discussed in [Sec. 3.2.4](#). Together, these subsections illustrate the progression from task-specific policy training to generalizable, semantically grounded control architectures.

3.2.1 Proximal Policy Optimization for Loco-Manipulation

Dynamic Stability Encoding in Reward Functions.

To prevent gradient suppression in high-dimensional action spaces, reward decomposition strategies isolate conflicting locomotion and manipulation objectives into distinct value functions [61, 96, 160]. These approaches employ semantic reward grouping with separate critic networks, utilizing normalized advantage summation or scheduled advantage-mixing to balance grasp stability terms against slip suppression metrics without policy collapse. Physical balance constraints integrate directly into the reward landscape through Zero Moment Point formulations and angular momentum regularization [63, 175]. This encoding penalizes deviations via non-planar ZMP calculations and inertial counter-torque generation, often leveraging dynamic tail utilization to maintain base orientation under load.

Entropy-Regularized Exploration in Contact-Rich Phases.

To prevent policy collapse during complex contact interactions, exploration mechanisms prioritize state-space perturbation and entropy maintenance over reward-based feasibility penalties. Sequential contact stage decomposition utilizes count-based curiosity rewards to drive task-agnostic exploration in long-horizon manipulation, incentivizing contact phase progression through dense stage counts rather than random network distillation [191]. Training sta-

Table 2 Taxonomy of representative methods in **End-to-End Learning-Based Control**. This table presents a taxonomy of end-to-end control methods, shows architectural hierarchies and knowledge origins, and highlights the prevalence of hierarchical designs leveraging motion corpora.

Method	Venue	Control Schema	Data Source	Highlight
Proximal Policy Optimization for Loco-Manipulation (Sec. 3.2.1)				
Whole-Body Force Policy [128]	ICRA’24	Force-Impedance RL	Proprioceptive Sensing	Soft contact potential fields enable force tracking via proprioception without explicit force-torque sensors.
WoCoCo [191]	CoRL’24	Stage-Based Sequential ZMP-Guided	Contact Sequences Privileged	Dense contact rewards and stage-count incentives mitigate exploration stagnation in multi-phase manipulation tasks.
DBHL [175]	IROS’25	Asymmetric Simulation	Simulation Simulated	ZMP-driven rewards and reward vectorization stabilize whole-body dynamics using only proprioceptive observations.
SoFTA [96]	arXiv’25	Frequency Decoupling	Randomization Explicit	Temporal decoupling of control frequencies prevents objective interference between locomotion stability and manipulation precision.
PFG [60]	IROS’25	Kinematic-Guided RL	Kinematics	Physical Feasibility-Guided rewards enforce kinematic reachability constraints to prevent convergence to suboptimal policies.
Adversarial Motion Priors & Imitation (Sec. 3.2.2)				
Adversarial Motion Priors [194]	IROS’24	Adversarial Imitation	Human MoCap	Integrates periodic phase rewards with AMP to stabilize gait generation while matching human motion distributions.
FB-CPR [156]	ICLR’25	Unsupervised Regularization	Unlabeled AMASS	Employs latent-conditional discriminators within Forward-Backward representations to align unsupervised policies with unlabeled trajectory distributions.
CWI [48]	RA-L’26	Dual-Discriminator IL	AMASS Expert Clips	Utilizes dual AMP discriminators and multi-critic PPO to enforce distinct locomotion style priors alongside manipulation tasks.
Vision-Language-Action Foundation Models (Sec. 3.2.3)				
ODYSSEY [166]	AAAI’25	Vision-Language Planning Vector	RGB-D LiDAR	Fuses RGB-LiDAR semantic graphs with terrain-invariant sampling for hierarchical whole-body control.
SONIC [107]	arXiv’25	Quantized Tokens	Large-Scale MoCap	Unifies heterogeneous motion commands via vector quantization and masked token prediction for cross-embodiment control.
PhysiFlow [132]	arXiv’26	Multi-Brain Decoupled VLA Tracking	Synthetic MoCap	Decouples semantic reasoning from physical execution using latent flow matching for high-frequency smooth motion.
OpenHLM [62]	arXiv’26	Hierarchy Async	Heterogeneous Teleop	Adapts VLA backbones to humanoid kinematics through joint-space retargeting and heterogeneous co-training strategies.
ReactiveBFM [20]	arXiv’26	Replanning	Text-Motion Corpus	Mitigates autoregressive exposure bias via scheduled prefix sampling and asynchronous replanning with temporal ensembling.
Hierarchical Skill Composition (Sec. 3.2.4)				
Puppeteer [55]	ICLR’24	Model-Based Hierarchy	CMU MoCap	Decouples motion tracking from task execution via reusable low-level world model and high-level geometric command generation.
SkillBlender [81]	arXiv’25	Skill Blending	Primitive Library	Blends frozen primitive skills using softmax-normalized per-joint weight vectors to ensure feasible whole-body coordination.
BAT [6]	arXiv’26	Hierarchical Switching	AMASS Motion	Aligns discrete motion tokens with controller regimes via option-aware VQ-VAE for robust high-level switching policies.
HANDOFF [184]	arXiv’26	Multi-Teacher Distillation	Retargeted Motion	Routes multi-teacher distillation through context-conditioned gating to reconcile conflicting locomotion and manipulation objectives.
CoorDex [95]	arXiv’26	Residual Latent Control	Simulated Demonstrations	Factorizes body-hand priors with coordinated latent residuals to adapt dexterity while preserving frozen mobility bases.

bility relies on time-varying action transformations that restrict the effective action space during early phases, employing scaling functions to reduce gradient variance before expanding to full torque limits [98]. Adaptive curricula further modulate task difficulty parameters such as target distance and stability thresholds to navigate local minima arising from the trade-off between aggressive dynamic motion and postural control [115]. These strategies collectively sustain behavioral diversity without relying on physical feasibility-guided reward terms that penalize kinematically unreachable configurations [60].

Proprioceptive Resilience via Observation Perturbation. Achieving sensor noise resilience requires injecting temporally correlated perturbations into

proprioceptive streams, often utilizing IMU-centric Ornstein-Uhlenbeck processes to approximate real-world jitter within high-frequency disturbance curricula [36, 197]. Zero-shot transfer fidelity further depends on neural actuator modeling to capture non-linear joint dynamics, ensuring the policy internalizes hardware-specific friction and backlash rather than relying on idealized simulation states [36, 128]. To decouple policy execution from specific physics engines, training frameworks emulate robot interfaces and estimate contact states, such as torque-based foot detection, without relying on physical force-torque sensors [103, 128]. These strategies collectively bridge the observation gap by forcing the network to learn invariant features against simulated sensor drift and

latency, thereby mitigating base shaking through relaxed tracking constraints derived from noisy inputs [103, 197].

3.2.2 Adversarial Motion Priors & Imitation

Adversarial Distribution Matching for Multi-Modal Motion Synthesis. To resolve kinematic ambiguities in high-dimensional state spaces, adversarial discriminators enforce distributional alignment between policy-generated trajectories and reference motion corpora through latent-conditional log-ratio estimation and forward-backward representations [48, 156]. Optimization conflicts arising from imbalanced motion library distributions are mitigated by decoupling locomotion and manipulation data sources via role-specific discriminators, where dual AMP critics isolate value heads to maintain distinct priors for walking versus squatting styles within filtered AMASS subsets. Separately, stable policy convergence when tracking imperfect video-derived references containing noisy contact information is achieved through residual action grounding and unified interaction rewards that utilize phase variables and local frame object poses to promote valid contacts during monocular video retargeting [174]. Furthermore, mapping heterogeneous motion modalities into a shared 64-dimensional latent command space facilitates single-policy imitation without explicit inverse kinematics solutions by employing modality-specific autoencoders for SMPL-X poses and end-effector SE(3) trajectories under an adaptive sampling curriculum [106].

Physics-Aware Discriminator Conditioning and Feasibility Filtering. Adversarial priors enforce physical feasibility by conditioning discriminators on contact states and joint limits rather than injecting diffusion-based regularization into the generative loop [64]. This mechanism explicitly suppresses foot-sliding and enforces kinematic constraints through latent representations that distinguish valid dynamics from impossible poses, ensuring policy outputs remain within executable bounds. To prevent collapse when imitating unstable or high-dynamic reference sequences, pre-training data filtering employs stability metrics based on the distance between the Center of Mass and Center of Pressure alongside adaptive reward scaling [176, 194]. Such frameworks treat complex balance requirements as strict feasibility filters rather than soft optimization objectives, utilizing periodic phase rewards modeled via angular distributions to maintain equilibrium during aggressive maneuvers without bi-level solvers. Distinct from these hard filters, probabilistic sampling and region-specific conditioning soften adversarial constraints

to resolve conflicts between motion fidelity and task success [136]. By applying Gaussian sampling around transitions and employing position-conditioned discriminators with region-specific motion buffers, these methods balance task-motion objectives while preserving dynamic plausibility.

Phase-Conditioned Discriminators and Partial Observability Resolution. Enforcing strict adherence to nonholonomic dynamics during underactuated transitions requires discriminators conditioned on explicit phase variables and kinematic coupling constraints [54]. This approach integrates board tilt-truck steering kinematics with phase-wise discrimination, utilizing Bezier curve trajectory guidance and slerp interpolation to manage contact phase switching without violating dynamic feasibility. Alternatively, hybrid state-space constructions fuse probabilistic torso sampling with model-based lower-body references to validate out-of-distribution whole-body configurations while avoiding kinematic bias [92]. Such frameworks enable the synthesis of hybrid motions by combining MoCap arm data with sampled torso commands derived from multi-contact optimal control, where teacher-student distillation functions strictly as a mechanism to resolve partial observability within the discriminator input space through proprioceptive history encoding, rather than facilitating high-level skill transfer or semantic grounding.

Morphological Divergence and Embodiment Gap. Static human motion corpora fail to cover the reachable workspace volumes of non-anthropomorphic morphologies, leading to kinematic singularities when policies attempt to track unreachable reference poses. Direct imitation on platforms with axial rotation waists or altered limb proportions induces severe tracking errors because standard datasets lack the necessary yaw rate dynamics and joint configurations [131]. These structural mismatches render blind locomotion setups ineffective for behaviors like straight-knee walking or heel-to-toe running without extensive manual re-targeting to align trajectories with specific embodiment constraints [194]. Consequently, reliance on fixed priors causes policy collapse in out-of-distribution regions where human biomechanics do not map to robot degrees of freedom, necessitating iterative co-optimization loops that adapt the adversary’s target distribution or the policy’s objective function to accommodate morphological mismatches rather than depending solely on pre-processed motion fusion [92].

3.2.3 Vision-Language-Action Foundation Models

Architectural Decoupling of Planning and Execution. Isolating high-level semantic planning from low-level physical execution manages whole-body complexity while preserving stability during locomanipulation transitions. Recent frameworks achieve this separation by encoding typed JSON subtasks and enforcing geometric feasibility gating to coordinate continuous motor commands without compromising base mobility [122, 137]. Bridging the sim-to-real gap further requires explicit verification loops where predicate-based success conditions trigger targeted recovery operators, allowing joint fine-tuning strategies to backpropagate tracking errors for zero-shot transfer without retraining the semantic planner [132, 137].

Latent Space Alignment for Bridging Semantic-Physical Gaps. Structured variational latent spaces align high-level semantic embeddings with physical action distributions, enabling smooth interpolation between unseen motion primitives through mechanisms such as conditional variational autoencoders that integrate CLIP text features and proprioceptive history [149, 183]. Vector quantization further discretizes these continuous trajectories to eliminate jitter while preserving diagonal walking patterns. To address embodiment discrepancies within the model architecture itself, latent alignment techniques map heterogeneous kinematic structures directly into a shared command space, unifying diverse robot morphologies through end-to-end representation learning [62, 204]. This architectural approach facilitates scalable data collection by coupling LiDAR base tracking with precise layer-specific fine-tuning on action layers. Heterogeneous co-training strategies subsequently extend semantic coverage by integrating stationary manipulation data with robot-free human motion sequences, utilizing hybrid data integration pipelines to preserve bimanual priors [62, 204]. While mixing sparse real-robot data with the HuMI dataset enhances dynamic balance, reliance on action mean squared error correlations remains a noted failure mode when generalizing to novel physical interactions.

Temporal Consistency Mechanisms. Scheduled curriculum sampling on noisy prefixes forces autoregressive planners to learn error-recovery capabilities from imperfect physical states, directly mitigating cumulative exposure bias [20]. To resolve latency mismatches between low-frequency generation and high-frequency control loops, asynchronous execution frameworks employ trajectory chunking and temporal ensembling, effectively eliminating inference-induced

jitter while decoupling synthesis rates such as 10 Hz planning from 50 Hz actuation [20, 132]. Replacing discrete decoding with continuous latent flow matching enables high-frequency smooth motion synthesis, maintaining physics-aware consistency through direct inference-time alignment of latent vectors without retraining loops. Furthermore, unified token spaces utilizing masked prediction for autoregressive in-betweening ensure temporal coherence across heterogeneous motion commands without requiring manual reward engineering or explicit cycle consistency losses [107].

Geometric and Dynamic Grounding. To suppress the generation of dynamically inconsistent or hallucinated motor commands, recent architectures inject physically feasible motion priors directly into the training objective using optimal control-derived synthetic datasets. By employing Differential Dynamic Programming to solve multi-phase optimal control problems, frameworks such as GR00TN1.5-3B align flow-matching objectives with energy-efficient trajectories that enforce strict adherence to torque limits [14]. Complementing these dynamic constraints, explicit geometric grounding restricts semantic planning within verified physical boundaries through 3D predicate evaluation and terrain-invariant sampling. Methods utilizing SAM3 for centroid estimation combine typed JSON preconditions with grasping logic aligned to principal object axes to ensure surface-normal compliance, while RGB-D depth gating triggers internal recovery mechanisms when perception-induced uncertainties threaten action feasibility [137, 166].

3.2.4 Hierarchical Skill Composition

Task-Frame Interfaces and Temporal Grounding. Hierarchical decomposition isolates manipulation intent from embodiment dynamics by defining skill boundaries through task-frame trajectory interfaces rather than body-relative commands [52]. This architectural choice enables camera-frame end-effector trajectories to execute asynchronously at distinct frequencies, such as a 5Hz policy loop coupled with a 50Hz controller, facilitating zero-shot cross-embodiment transfer and decoupled perturbation compensation. To segment continuous sensory streams into discrete primitives, temporal grounding mechanisms employ fingerprinting offsets or model-based trajectory prediction to trigger specific low-level policies [110, 153]. Techniques ranging from audio constellation-map fingerprinting with 5s chunk segmentation to hybrid ball dynamics estimation for racket target generation ensure robust temporal alignment. Intermediate transition states and unified orchestration interfaces further mediate skill

switching to maintain dynamic stability during high-frequency policy selection in non-stationary environments [110, 153]. By utilizing stand-priming latency buffers and TCP dispatch protocols, these systems satisfy sub-second reaction constraints while respecting choreography granularity limits and activating stability-triggered fallbacks when necessary.

Latent Command Unification and Kinematic Decoupling. Heterogeneous motion references, ranging from SMPL-X poses to end-effector SE(3) trajectories, converge into a shared 64-dimensional latent command space through modality-specific autoencoders, thereby bypassing explicit inverse kinematics preprocessing [106]. This representation learning facilitates asymmetric actor-critic architectures that align implicit commands without requiring geometric solvers. Complementing this unification, kinematic normalizing flows model multimodal inverse kinematic distributions to separate redundancy exploration from dynamic execution constraints [58]. Such flows enable latent space navigation across torso-arm configuration distributions while applying feasibility filtering to ensure physical validity. Reusable low-level world models further abstract this process by translating frozen geometric targets into whole-body actions, effectively decoupling motion tracking fidelity from high-level planning objectives [55]. These trackers, often pretrained on MoCap data, predict termination probabilities to guide MPPI planners without retraining the underlying controller. Finally, regularized residual adaptation mechanisms permit rapid specialization to novel motion distributions while maintaining the stability of these frozen base policies [173]. By enforcing Parseval-guided orthogonality and KL-divergence constraints within a Mixture-of-Experts backbone, these methods mitigate distributional drift using Center-of-Mass observations.

Library Expansion via Rollout-Based Data Aggregation and Residual Adaptation. Iterative rollout-based data aggregation mitigates distribution shifts induced by skill composition without requiring architectural modifications to the underlying controller. By augmenting closed-loop execution with expanded reference trajectory sets and domain randomization, these methods maintain a task-agnostic low-level interface that remains robust against skill-induced state shifts [78]. Cross-embodiment transfer further decouples structural kinematic mismatches from dynamic residuals through deterministic alignment matrices and frozen motion priors. This approach utilizes sparse scattering matrices and hip decoupling blocks to handle parallel joint coupling, while low-rank LoRA injection adapts only the proprioception-action head to new morphologies [185]. For coordinated loco-

manipulation, asymmetric prior factorization employs frozen body and hand teachers to guide a shared latent residual policy. Such teacher-student distillation enforces wrist-finger separation and distinct residual heads, enabling continuous locomotion and grasping through a shared coordination trunk [95].

Conflict Resolution via Dynamic Blending. To resolve objective conflicts during simultaneous execution, context-conditioned multi-teacher distillation dynamically routes supervision to specific body segments based on runtime regime signals such as velocity magnitude or recovery status [6, 184]. This approach employs action-sliced KL divergence and binary fall-recovery masks to guide confidence-weighted decision fusion, ensuring that distributional shifts do not compromise stability during sliding-horizon pre-evaluation. Complementing this routing strategy, vectorized per-joint weighting mechanisms synthesize complex whole-body behaviors by applying non-linear normalization to frozen primitive skill outputs [81]. By utilizing softmax-normalized weight vectors in a Hadamard product fusion, these systems prevent reward hacking while enforcing tilt and torque feasibility constraints for subgoal-driven blending.

3.2.5 Discussion

Architectural shifts toward hierarchical and residual frameworks effectively decouple high-level semantic intent from low-level physical stability, thereby mitigating the expressiveness-safety trade-off inherent in monolithic policies through mechanisms such as scheduled advantage-mixing and asymmetric body-hand factorization [61, 95, 186, 195]. While foundation model integration alleviates data efficiency bottlenecks by transferring knowledge from large-scale multimodal corpora rather than relying on exhaustive on-policy exploration, this approach introduces latency constraints, with flow matching inference times reaching 18.65 ms, which challenges real-time control loop requirements [132, 180, 182]. Furthermore, although techniques like teacher-student distillation and Extended Random Force Injection (ERFI) help bridge observation gaps for hardware deployment, the reliance on probabilistic convergence rather than formal Lyapunov stability proofs means absolute safety guarantees remain compromised, particularly when managing Series Elastic Actuator nonlinearities and action delay stochasticity [102, 182, 186, 195].

3.3 Hybrid Model-Learning Integration

Hybrid model-learning integration architectures seek to unify the theoretical guarantees of model-based control with the adaptive capacity of data-driven

methods. [Sec. 3.3.1](#) investigates residual reinforcement learning frameworks that refine baseline controllers by learning corrective actions within a constrained space. [Sec. 3.3.2](#) examines learning-augmented dynamics models where neural networks compensate for unmodeled disturbances inside predictive horizons. To enhance sampling efficiency, [Sec. 3.3.3](#) details approaches using learned policies to guide exploration distributions in sampling-based MPC. Finally, [Sec. 3.3.4](#) addresses safety-critical filtering layers that enforce constraint satisfaction on learned outputs through minimal intervention mechanisms.

3.3.1 Residual Reinforcement Learning

Architectural Decoupling and Baseline Design Strategies. Control authority is decomposed by assigning a model-based optimizer to generate nominal feedforward trajectories, such as quadratic program torques or additive acceleration residuals, while restricting the learned policy to low-frequency feedback corrections for reactive disturbance rejection [24, 109]. This hierarchical separation further isolates high-level visual manipulation intents from low-level whole-body dynamics, enabling asynchronous execution rates where a 5 Hz diffusion policy commands task-frame trajectories for a 50 Hz joint tracker [52]. Such architectural constraints effectively limit the reinforcement learning exploration space to unmodeled dynamics and contact errors rather than full state-space stabilization, facilitating privileged state distillation and payload generalization with reduced action dimensionality [24, 109].

Baseline Stability and Exploration Regularization. Hierarchical optimization frameworks utilize a model-based follower to maintain nominal tracking performance, allowing the learned policy to focus exclusively on correcting unmodeled dynamic errors rather than enforcing fundamental stability [163]. This architectural decoupling isolates complex kinematic interactions and contact dynamics from the adaptive layer, effectively mitigating the sim-to-real gap by separating rigid physical priors from flexible behavioral tuning. Complementing this structural separation, privileged model-based teachers stabilize reinforcement learning exploration by integrating supervision terms directly into the objective function [162]. By modifying the PPO objective with these guidance signals during the actor-critic warm-up phase, the system constrains the policy search space around feasible trajectories, ensuring a controlled transition from imitation to autonomous control even under chassis noise injection.

Gradient-Based Credit Separation in Residual Loops. Hybrid control frameworks distinguish policy contributions by computing gradients exclusively for the residual term while treating the baseline controller as a fixed reference during backpropagation. This additive composition allows the system to attribute performance gains directly to learned corrections for unmodeled dynamics, such as friction compensation and motor strength alignment, without altering the nominal trajectory generator [57]. By restricting updates to the delta action model based on real-world rollout discrepancies, the optimization process isolates the credit for disturbance rejection from the baseline feedforward commands [66]. Such mathematical separation ensures that the learning signal reflects only the efficacy of the corrective layer in minimizing state mismatch, thereby preventing the conflation of baseline stability with adaptive refinement.

Convergence Dynamics and Sample Efficiency Gains. Residual formulations accelerate policy convergence by restricting the optimization landscape to local corrections rather than global stabilization, yielding faster sample efficiency compared to training from scratch [195]. By modeling full return distributions under perturbed dynamics, these frameworks capture environmental uncertainty more effectively than standard critics, stabilizing the learning curve during early training phases. This structural constraint reduces the effective action dimensionality, allowing agents to adapt to new tasks with minimal real-world interaction while maintaining the stability of the nominal gait [65]. Consequently, the learning process achieves rapid fine-tuning driven by high-level planning signals without requiring extensive exploration of the full state space. Such focused updates ensure that convergence rates remain high even under dynamic uncertainties, as the policy refines only the delta actions necessary to minimize state mismatch [44].

3.3.2 Learning-Augmented Dynamics Models

Residual Dynamics Integration in MPC Constraints. Integrating analytical physics backbones with neural networks enables the explicit modeling of unmodeled disturbance terms, effectively correcting prediction errors across the MPC horizon [66, 69, 120]. Such frameworks embed learned residuals directly into the state propagation constraint, typically formulated as $x_{k+1} = f(x_k) + g(x_k, u_k) + \hat{d}_\theta$, where the neural term $\hat{d}_\theta$ compensates for discrepancies between the nominal model and actual system behavior. By inserting this additive term directly into the dy-

Table 3 Taxonomy of representative methods in **Hybrid Model-Learning Integration**. This table presents a taxonomy of hybrid control methods, shows distinct fusion strategies and safety mechanisms, and highlights the trade-off between adaptability and formal guarantees.

Method	Venue	Fusion Strategy	Safety Mechanism	Highlight
Residual Reinforcement Learning (Sec. 3.3.1)				
HiLma-Res [65]	IROS'24	Residual Learning	Hierarchical Decoupling	Integrates Bézier residual curves with CPG gaits to enable stable, task-specific foot placement adjustments. Learns delta action models from real data to correct simulation dynamics mismatches during policy fine-tuning. Applies RL feedback corrections to model-based WBC references, reducing exploration while preserving force tracking precision.
ASAP [57]	Robotics'25	Delta Correction	Sim Alignment	
RAMBO [24]	IEEE'25	Feedback Correction	QP Constraints	
Learning-Augmented Dynamics Models (Sec. 3.3.2)				
WTNPB [72]	ICRA'24	Parametric Bias	Implicit Adaptation	Parametric Bias vectors implicitly encode tool properties, enabling online adaptation of sensorimotor dynamics without retraining. Cosserat rod priors integrated into Augmented Neural ODEs correct shape estimation mismatches for robust MPC planning.
Shape-Control-NODE [68]	ICRA'25	Neural ODE Coupling	Physics Priors	
RL-Guided Sampling MPC (Sec. 3.3.3)				
Whole-Body MPPI [3]	ICRA'25	Sampling-Based MPC	Collision Costs	Spline parameterization compresses action space, enabling tractable whole-body sampling without RL priors. Hierarchical MPC steers low-level policy commands, reducing dimensionality while preserving safety overrides. RL policy means bias Gaussian sampling, concentrating exploration in dynamically feasible regions.
Sumo [193]	arXiv'26	Policy-in-Loop	Reduced Command Space	
RGB [147]	arXiv'26	RL-Guided Sampling	Cost Constraints	
Safety-Critical Filtering Layers (Sec. 3.3.4)				
A-ECBF QP Controller [123]	ACC'24	Inverse Dynamics QP	Exponential CBFs	Formulates acceleration-based ECBFs within inverse dynamics QP to ensure safety without mass matrix inversion. Applies CBFs at the initial timestep for exact self-collision avoidance within truncated ADMM-based NMPC. Integrates ISSf-CBF filters on reduced-order models to robustly transfer kinematic safety guarantees to full-body dynamics.
Low-Acc Whole-Body NMPC [75]	IEEE'24	Single-Layer NMPC	Control Barrier Functions	
ISSf-CBF Safety Filter [85]	arXiv'26	Hierarchical Filtering	ISSf-CBF Layer	

namics equation rather than treating it as a separate policy correction, the optimizer computes gradients through the learned network during each iteration of the solution process. This direct coupling ensures that the control sequence accounts for predicted disturbances while maintaining the structural integrity of the underlying physical model, allowing controllers to address complex phenomena like aerodynamic spin or soft-body deformation without retraining the entire policy [66].

Latent Parameterization for Online Adaptation.

Embedding unmodeled physical variations into low-dimensional bias vectors enables a single network to represent multiple distinct dynamic regimes without explicit retraining [71, 72]. This parametric bias p facilitates implicit self-organization within the latent space, allowing the model to cluster representations of varying tool properties, body state changes, or footwear adaptations. Online adaptation occurs exclusively through the optimization of these latent variables against immediate sensor feedback, decoupling rapid inference from computationally expensive weight updates [71, 72]. By performing gradient descent on the bias terms to minimize errors such as Zero Moment Point (ZMP) deviations or muscle

tension irregularities, the system achieves real-time fine-tuning while preserving accurate modality reconstruction.

Physics-Informed Operators and Constraint Tightening.

Embedding analytical continuum mechanics, such as Cosserat rod theory, directly into the backbone of augmented neural ordinary differential equations ensures that gradient propagation adheres to structural conservation laws during continuous shape state estimation [68]. This architectural constraint mitigates model mismatch by aligning the learned dynamics with physical priors, thereby minimizing trajectory errors in deformable systems. Complementing this structural fidelity, integrating neural network-enhanced state covariance predictions into Kalman filtering mechanisms dynamically adjusts uncertainty estimates to bridge the simulation-to-reality gap for accurate velocity tracking [146]. Such adaptive uncertainty quantification explicitly tightens constraints within the MPC optimization horizon, employing techniques like tube MPC or chance constraints to define robust feasible sets that account for real-world disturbances. Coupling these physics-constrained estimators with closed-loop MPC policies enables the simultaneous optimization of whole-body

trajectories while maintaining stability under external payload variations [68, 146]. The resulting framework utilizes joint training of the dynamics predictor and action regularization to generalize across diverse payloads without compromising shape-aware control performance within the optimization horizon.

3.3.3 RL-Guided Sampling MPC

Prior-Centered Sampling Mechanisms. By defining the mean of the sampling distribution through learned policies, controllers effectively constrain stochastic exploration to regions of high dynamic feasibility [147]. This Gaussian centering around pretrained outputs suppresses trajectory drift and concentrates computational resources on viable control sequences. Extending this concept, hierarchical architectures employ low-level stabilized policies as rollout engines, allowing high-level planners to optimize within a compressed command space rather than raw actuator dimensions [193]. Such policy-in-the-loop rollouts facilitate stable planning for contact-rich tasks by targeting abstract variables like torso velocity or arm endpoints. Crucially, decoupling prior generation from cost function evaluation enables online task adaptation through modular objective updates without requiring policy retraining [147, 193]. This separation maintains fixed interface robustness during receding-horizon refinement while accommodating diverse objectives through updated cost terms.

Covariance and Temperature Modulation. Distinct from mean-centering strategies, learned policies modulate the covariance structure and temperature parameters to regulate sampling entropy [3]. By dynamically scaling the variance of the Gaussian distribution, these methods adjust the breadth of stochastic rollouts independent of the trajectory centroid [104]. This mechanism permits aggressive exploration during high-uncertainty phases while contracting the search volume near convergence, effectively tuning the exploration-exploitation balance without shifting the distribution mean. Consequently, the optimizer refines trajectories with reduced sample budgets by adapting the noise profile to local cost landscape curvature rather than relying on fixed temperature schedules [3, 104]. Such adaptive scaling enables robust performance in contact-rich scenarios by maintaining sufficient diversity to escape local minima while preventing excessive deviation from feasible control manifolds.

Computational Latency in Parallel Inference. Integrating learned priors into sampling-based MPC introduces a distinct computational burden by requiring parallel neural network inference alongside

thousands of trajectory rollouts per control cycle [188]. While high-level policies effectively reduce the search dimensionality by specifying abstract targets like forward velocity and contact locations, the latency incurred by evaluating these networks can constrain the maximum feasible sample count or horizon length. This architecture creates a direct trade-off between the complexity of the guidance policy and the number of samples available for statistical convergence within fixed time steps. Safety enforcement remains the responsibility of the cost function or external modules, avoiding the need to detail specific override mechanisms here. Consequently, system design must balance the informational gain from sophisticated priors against the reduction in sampling density caused by inference overhead.

3.3.4 Safety-Critical Filtering Layers

High-Relative-Degree Formulations for Whole-Body Dynamics. To enforce safety constraints with high relative degrees, recent approaches transform collision avoidance conditions directly into acceleration-level linear inequalities within whole-body control optimizations [74, 123]. By utilizing exponential Control Barrier Functions tailored for relative-degree-two systems, these methods incorporate signed distance functions and joint accelerations as explicit variables in quadratic program inequality constraints. Reformulating barrier functions to depend directly on joint accelerations eliminates the computational burden of explicit mass matrix inversion while preserving the forward invariance of safe sets [123]. This Acceleration-based Exponential CBF (A-ECBF) formulation integrates naturally with inverse dynamics quadratic programs, supporting closed-chain kinematics and zero moment point constraints with improved numerical efficiency.

Ensuring Optimization Feasibility via Hierarchical and Hybrid Constraints. To maintain solver feasibility under strict computational budgets, hybrid architectures often enforce rigid Control Barrier Functions only at the initial timestep while relaxing non-linear kinematic constraints over the remaining prediction horizon [75]. This approach typically pairs single-step Sequential Quadratic Programming with truncated Alternating Direction Method of Multiplier iterations to balance modeling fidelity against available processing time. Distinct from temporal relaxation, multi-tier filtering stacks decouple kinematic reference retargeting from dynamic torque enforcement to mitigate policy overshoot and guarantee contact-consistent stability [113]. Such hierarchies utilize inverse kinematics for high-level trajectory adjustment followed by joint torque quadratic programs

that integrate task-consistent weighting to preserve forward invariance. When reactive acceleration-level barriers encounter infeasibility during extreme disturbances, purely local constraint satisfaction proves insufficient without heuristic augmentation [74]. Integrating swing-leg reflexes alongside signed distance function inequalities within a Linear Model Predictive Control framework addresses these convexity limitations, enabling recovery stability that static barrier formulations cannot achieve alone.

Adaptive Tuning via Energy Tanks and Dynamic Bounds. To mitigate the conservativeness of fixed uncertainty margins, recent methods replace static disturbance bounds with adaptive energy tank mechanisms that dynamically modulate admittance parameters [9]. By treating the system as an energy storage function, these approaches decouple passivity constraints from real-time adaptation, regulating power flow through convex quadratic programming to accommodate variable interactions in human-robot collaboration without compromising stability. This dynamic modulation contrasts with Input-to-State Stable frameworks that rely on scaling conservative tuning parameters against pre-defined model mismatch terms [85]. Optimal filtering performance thus depends on balancing these adaptive energy limits against task execution fidelity, ensuring that collision ratio minimization and operator force reduction do not result in excessive safety margins that degrade operational efficiency or fail to tolerate mass discrepancies [9, 85].

Policy Failure Recovery via Latent Fault Estimation. When nominal policies persistently request actions that violate physical limits, latent fault estimation remaps unsafe center-of-mass commands into feasible support boundaries to prevent solver divergence in whole-body optimizations [205]. This approach utilizes a proprioceptive history encoder to identify degraded support polygons caused by actuator failures, applying a non-linear mapping via a Posture Adaptation Module to preserve stability. By intervening at the command level prior to optimization, this mechanism addresses failure modes where the learned policy itself generates infeasible trajectories, distinct from reactive barrier formulations that struggle with local constraint satisfaction during extreme disturbances.

3.3.5 Discussion

Embedding differentiable learning modules within rigorous optimization horizons effectively resolves the safety-adaptability dichotomy by ensuring forward invariance while accommodating unmodeled dynam-

ics [23, 120, 162, 192]. Hierarchical decoupling of high-level sampling from low-level execution further stabilizes the search space, enabling zero-shot generalization through standardized command interfaces that separate global planning from local trajectory refinement [104, 152, 193]. To maintain these guarantees without stifling exploratory capacity, integration frameworks increasingly rely on Control Barrier Functions and adaptive laws that enforce constraints via minimal intervention [123, 152, 192]. However, tuning the interaction between model-based guarantees and data-driven flexibility remains a non-trivial challenge, as balancing reward sparsity against computational overhead often exposes vulnerabilities in sim-to-real transfer and velocity tracking [23, 104, 120].

3.4 Human-in-the-Loop & Teleoperation Control

Human-in-the-loop control frameworks integrate operator intent with robotic execution to address complex manipulation tasks through diverse interaction modalities. Sec. 3.4.1 establishes the foundational kinematic retargeting algorithms required to map human motions onto robot morphologies while preserving stability and respecting joint limits. Building on this motion mapping, Sec. 3.4.2 examines bilateral teleoperation systems that transmit force feedback to enhance situational awareness during delicate physical interactions. To further improve performance, Sec. 3.4.3 explores shared autonomy architectures where predictive models blend human commands with automated safety constraints and tracking behaviors. Finally, Sec. 3.4.4 extends these concepts to robot-free demonstration learning, analyzing methods that bridge embodiment gaps to transfer whole-body policies captured without direct hardware presence.

3.4.1 Kinematic Retargeting & Motion Mapping

Deterministic Optimization for Singularity Resolution and Morphological Retargeting. Formulating retargeting as a constrained nonlinear problem allows systems to prioritize incremental configuration changes over absolute state mapping, ensuring solution existence near kinematic limits through single-iteration sequential quadratic programming and decision variable dimensionality reduction [143, 144]. By strictly adhering to deterministic optimization frameworks, these methods resolve morphological mismatches by minimizing joint displacement costs while satisfying hard kinematic constraints, thereby avoiding the numerical instability associated with singular configurations without relying on compliant

Table 4 Taxonomy of representative methods in **Human-in-the-Loop & Teleoperation Control**. This table presents a taxonomy of teleoperation frameworks, shows distinct control strategies and data origins, and highlights the shift toward hierarchical learning architectures.

Method	Venue	Control Paradigm	Data Source	Highlight
Kinematic Retargeting & Motion Mapping (Sec. 3.4.1)				
SEIKO [143]	T-Mech’22	Optimization-Based SQP	Analytical Dynamics	Single-iteration SQP solves incremental kinematic changes under quasi-static equilibrium for real-time retargeting.
Human to Humanoid (H2O) [56]	IROS’24	Goal-Conditioned RL	Filtered AMASS Set	Privileged simulator filters dynamically infeasible motions to train robust whole-body tracking policies.
PMP [105]	ICRA’25	Decoupled RL-IK	Retargeted Motions	CVAE encodes future upper-body trajectories as latent priors to stabilize decoupled lower-body locomotion.
SMAP [200]	arXiv’25	Teacher-Student RL	CMU Mocap Dataset	Vector-quantized autoencoder maps heterogeneous human motion to a shared phase manifold for feasible imitation.
NMR [201]	arXiv’26	Supervised Transformer	SMPL Mocap Data	Non-autoregressive Transformer reformulates retargeting as distribution mapping to bypass non-convex optimization traps.
Bilateral Force Feedback & Haptics (Sec. 3.4.2)				
Purushottam et al. [129]	RA-L’23	Reduced Order Model	HMI Sensor Data	Delivers scaled interaction forces and virtual spring feedback to the human torso for immersive force regulation.
Hybrid Pos-Force Teleop [130]	Humanoids’25	Switched Impedance	HMI Moment Est	Renders whole-body environmental contact moments to the operator’s center of mass to enhance situational awareness.
CHILD [117]	Humanoids’25	Direct Kinematic Map	Encoder-IMU Fusion	Applies adaptive virtual spring torques via direct joint mapping to prevent singularities and guide limb return.
Shared Autonomy & Intent Prediction (Sec. 3.4.3)				
H2-COMPACT [12]	Humanoids’25	Diffusion-PPO Hybrid	Force-Torque Signals	Multiscale diffusion policy infers intent from wavelet-encoded haptic signals for robust hierarchical locomotion control.
X-OP [164]	arXiv’26	Hierarchical MPPI	Digital Twin Sim	MPPI optimization jointly aligns operator intent with dynamic feasibility while correcting simulator drift via state synchronization.
Coupled Ego Control [99]	arXiv’26	Rule-Based Coupling	Real-Time Poses	Dual coupling policies automate base and torso motions triggered by head orientation and end-effector workspace boundaries.
Robot-Free Demonstration Learning (Sec. 3.4.4)				
Moyen et al. [114]	Humanoids’25	Inverse Dynamics IK	Human Demonstrations	Compares decoupled versus coupled control strategies to assess embodiment gap impacts on imitation learning data quality.
HuMI [118]	arXiv’26	Diffusion-RL Hierarchy	Wearable Sensors	Integrates online kinematic visualization and relative pose tracking to ensure robot feasibility in robot-free whole-body demonstrations.
BifrostUMI [187]	arXiv’26	Diffusion Visuomotor	VR Keypoint Trajs	Proposes Spatial Keypoint Retargeting to map sparse human keypoints to robot-native motions while preserving metric task geometry.

mechanical models or probabilistic trajectory generators.

Learning-Based Distribution Mapping. Reformulating retargeting as a supervised distribution mapping problem bypasses the non-convex optimization landscapes and local optima inherent in frame-wise geometric solvers by utilizing CNN-Transformer backbones trained on pseudo-ground-truth data to enforce kinematic alignment and physical grounding [201]. To ensure policy robustness against morphological constraints prior to real-world deployment, privileged simulation filtering removes dynamically infeasible motion sequences from training datasets, enabling zero-shot transfer through goal-conditioned reinforcement learning with domain randomization [56]. Vector-quantized autoencoders further address heterogeneous action spaces by aligning human and robot movements into a shared discrete phase manifold, generating physically plausible imitation goals without paired supervision via teacher-student dis-

tillation [200]. Finally, mixture-of-experts architectures resolve conflicting whole-body control objectives by dynamically routing inputs to specialized sub-networks for distinct motion regimes, effectively integrating adversarial motion priors to manage locomotion-manipulation switching and drift correction within a unified policy network [97].

Latency Compensation via Predictive Motion Priors and Closed-Loop Feedback. To synchronize robot actions with operator perception despite communication delays, probabilistic movement primitives conditioned on delayed observations enable the execution of anticipated trajectories [125]. This approach utilizes posterior conditioning and minimum distance task recognition to sample trajectories ahead of command arrival, ensuring smooth policy blending during round-trip delays. Beyond temporal alignment, encoding predicted future upper-body motions into a latent vector provides a motion prior that stabilizes decoupled lower-body locomotion policies

against dynamic disturbances caused by unmodeled arm dynamics [105]. By augmenting policy observations with these conditional variational autoencoder embeddings, systems achieve adaptive disturbance rejection without requiring coupled inverse kinematics solutions. For long-horizon operations, integrating global pose discrepancy vectors derived from LiDAR odometry directly into the policy input creates a closed-loop feedback mechanism that suppresses unbounded positional drift through continuous error correction based on real-time position feedback.

Sensor-Driven Retargeting Strategies for Drift Mitigation and Real-Time Stability. To eliminate cumulative yaw drift inherent in wearable IMU suites during long-horizon operations, trigger-based relative motion mapping decouples absolute sensor orientation from task-space commands by switching velocity references upon discrete hand closure events [206]. Dedicated kinematic streaming toolboxes further compensate for network jitter and low-frequency user inputs through first-order prediction and bounding box filtering prior to inverse kinematics resolution, effectively upsampling 60 Hz data to meet 1 kHz control bandwidth requirements [11]. Physics-agnostic exponential moving average filters attenuate high-frequency sensor noise while preserving motion responsiveness without requiring domain randomization or dynamic model tuning, thereby enabling unified simulation-to-real execution with minimal phase lag [38].

3.4.2 Bilateral Force Feedback & Haptics

Passivity and Delay Compensation in Force Loops. Achieving stability within delayed bilateral control loops requires enforcing passivity constraints that prevent energy generation despite communication latency, rather than relying on kinematic trajectory abstraction. Methods utilizing wave variables or scattering theory transform force and velocity signals to preserve system passivity, ensuring stable haptic coupling even when time delays would otherwise induce oscillations [29, 30]. Alternative approaches employ energy tank architectures that monitor the net energy flow of the teleoperation system, dynamically damping force feedback commands whenever stored energy reserves approach depletion to guarantee bounded behavior. These stability guarantees are often realized through spring-like haptic couplings that enforce passivity directly within the force loop, maintaining safe interaction through high-backdrivability interfaces without generating energy despite signal latency [168].

Whole-Body Moment Rendering. Recent architec-

tures extend feedback beyond local end-effector forces by mapping estimated external contact moments directly to the operator’s center of mass to maintain situational awareness during occluded interactions [130]. This whole-body rendering integrates robot dynamics to convey volatile sliding estimates and torso-level torques that would otherwise remain undetected by hand-held interfaces. To address high-load manipulation tasks, virtual spring force integration scales external interaction forces, effectively extending the human leaning range and amplifying the effort required to break static friction [129]. Such force regulation within a Human-Machine Interface allows operators to modulate their posture against scaled feedback rather than raw environmental resistance.

Adaptive Gain and Perceptual Scaling. To enhance perceptual fidelity during mode transitions, virtual spring models apply adaptive bias torques that scale force feedback magnitude without dictating specific limb trajectories or nominal configurations [117]. Distinct from static mapping, dynamic gain modulation switches controller behavior between compliance and assertion based on interaction force thresholds, balancing safety with clear intention signaling in physical collaboration [17]. Furthermore, velocity-dependent force profiling employs time-derivative sigmoid functions to scale repulsive feedback intensity, ensuring haptic warnings convey constraint proximity directly to the operator rather than triggering autonomous trajectory compensation [5]. These mechanisms collectively adjust the perceptual fidelity of teleoperation by modulating how environmental constraints are rendered to the operator.

Dynamic State Ambiguity and Contact Instability. Estimating whole-body moment feedback often suffers from high volatility during intermittent contact phases, failing to provide stable cues for continuous planning specifically across sliding-to-sticking transitions or in occluded scenarios [130]. Such instability is further exacerbated when feedforward human moment terms introduce signal ambiguity within the haptic loop; rather than stabilizing the operator, these inputs can amplify natural balancing fluctuations to induce oscillatory robot yaw behaviors and unintended coupling. Furthermore, while haptic warning channels offer safety benefits in visually degraded or unknown settings, they introduce perceptual uncertainty in familiar environments that degrades task performance through visual-haptic segregation and increased completion times [5].

3.4.3 Shared Autonomy & Intent Prediction

Probabilistic Intent Inference via Motion Primitives and Latency Compensation. To counteract communication delays, shared autonomy architectures execute anticipated trajectories derived from probabilistic motion priors before explicit operator commands arrive [125]. These systems utilize Probabilistic Movement Primitives (ProMPs) to condition posterior distributions on partial observations, enabling trajectory sampling that synchronizes visual feedback with round-trip latency constraints. Complementing this predictive layer, optimization-based frameworks enforce dynamic feasibility and safety by jointly minimizing tracking errors and collision risks within a receding horizon [164]. Such approaches often employ Model Predictive Path Integral (MPPI) control to solve nonlinear least-squares state synchronization problems, where cost function weighting balances contact stability against simulator drift correction.

Orthogonal Subspace and Constraint Blending.

Orthogonal projection matrices isolate human-led motion axes from robot-stabilized degrees of freedom, preventing control conflict without requiring physical disconnection [50, 178]. This task-space decoupling manipulates reference inputs to maintain null-space impedance while preserving constraint-free subspace dynamics for the operator. When safety and tracking objectives compete, convex optimization frameworks resolve the arbitration by encoding operational limits as hard inequalities within a Quadratic Programming formulation [9, 178]. Such approaches linearize friction cones and torque limits to define convex feasible sets that guarantee adherence to physical boundaries.

Non-Haptic Command Smoothing via Diffusion Priors.

By modeling the non-linear mapping between noisy kinematic inputs and desired locomotion velocities, multiscale conditional diffusion policies effectively decode operator intent [12]. These architectures employ stationary wavelet transforms to encode partial trajectory observations and positional jitter, enabling probabilistic trajectory sampling that generates robust velocity commands despite sensor noise. Such approaches bypass rigid analytical models, instead learning complex distributions to stabilize motion assistance through direct interpretation of stochastic human inputs.

Context-Dependent Efficacy and Coordination Overhead.

Shared autonomy efficacy exhibits a sharp dichotomy where automated intervention degrades performance in familiar settings yet significantly reduces mental load and enhances safety in novel or visually degraded environments [5]. This

variance stems from the introduction of uncertainty via non-linear assistance scaling and obstacle repulsive forces, which conflict with established operator priors but provide critical guidance when visual feedback is insufficient. Coupling egocentric perception signals to automated base and torso adjustments further lowers cognitive demand by eliminating explicit coordination overhead during interaction-rich tasks [99]. Such perception-centered mechanisms utilize threshold triggering to detect workspace boundary breaches, thereby reducing button press frequency and maintaining task flow continuity without requiring manual intervention. In dynamic gait training scenarios, decoupling mobile base inertia through compliant 3-DoF pelvic interfaces enables transparent assist-as-needed balance support [171]. Arbitration logic in these systems activates braking mechanisms only during fall events, minimizing inertial force transmission while simultaneously improving step length and walking speed.

3.4.4 Robot-Free Demonstration Learning

Latent Space Abstraction for Morphology-Agnostic Coordination.

Mapping disparate partial-body inputs into a shared spherical latent manifold enables policy distillation that operates without fixed interface assumptions, effectively unifying arbitrary keypoint subsets through masked transformer attention [94]. To further decouple task-space geometric intent from embodiment-specific dynamics, explicit spatial retargeting modules transform sparse trajectories, such as five-keypoint sequences, into high-dimensional joint references while preserving metric relationships and scaling vertical height [187]. Complementing these structural approaches, embodiment-agnostic visual representations utilizing 3D geometry-aware tokens and body part masking mitigate viewpoint mismatches, allowing relaxed action abstractions like gripper-centric coordinate frames to bridge kinematic gaps in head and end-effector control [179].

Physics-Aware Regularization to Compensate for Absent Force Feedback.

Injecting physics-aware regularization terms directly into the diffusion training objective biases the generative model toward executable motion manifolds prior to any tracking stage [64]. Specific losses enforcing contact consistency and temporal smoothness suppress foot-sliding artifacts and reduce jitter in decoded joint trajectories, effectively embedding dynamic constraints without real-time haptic cues. Approaches such as Clustered-Expert Physics Refinement (CEPR) utilize parallel RL expert policies to generate pseudo-ground-truth data, ensuring self-collision avoidance and strict adherence to joint limits through a non-autoregressive

CNN-Transformer architecture [201]. These hierarchical data refinement pipelines encode dynamic constraints directly into the supervision signal, substituting absent force feedback with offline physical grounding [64, 201]. By employing cluster-and-distill strategies alongside morphology-specific data generation, such systems achieve support-foot stability and improved hardware tracking performance despite the lack of embodied interaction during demonstration capture.

Action Representation Strategies to Mitigate Distribution Shift. Formulating policies to predict delta end-effector poses within an egocentric reference frame significantly curbs error accumulation during long-horizon execution compared to absolute or joint-space formulations [46]. To resolve discontinuities arising from tracking latency and kinematic feasibility gaps, subsequent action chunks can be anchored to scheduled target poses rather than executed states [118]. While direct joint-space command generation via heterogeneous co-training preserves whole-body coordination semantics, this approach often struggles to transfer novel motion patterns without sufficient data scale [62].

Dynamic Feasibility Filtering. Decoupled control paradigms frequently generate kinematically valid trajectories that fail to maintain dynamic coherence, particularly when base-arm coupling fidelity is compromised. Comparative analysis indicates that without strict filtering of whole-body demonstration data, methods such as Separate Body Controllers (SBC) and Task Space Inverse Dynamics suffer from high imitation learning failure rates due to an unaddressed base-arm coordination gap [114]. Consequently, enforcing dynamic consistency constraints during dataset curation becomes essential to prevent the propagation of physically unrealizable motions, a necessity highlighted by observed complete failure in unfiltered Whole Body Controller (WBC) evaluations.

3.4.5 Discussion

The fundamental bottleneck in whole-body teleoperation remains the limited bandwidth of human operators to process high-frequency dynamic states, necessitating a shift from direct command mapping to predictive intent inference [12, 125]. Effective collaboration addresses this constraint by abstracting complex humanoid dynamics into reduced-order models that preserve stability margins while allowing humans to supervise via high-level kinematic or force references rather than low-level joint torques [29, 30, 178]. While hierarchical autonomy and robot-free demon-

stration learning decouple style transfer from control stability to reduce cognitive load [64, 137, 201], teleoperation will likely persist primarily as a critical fallback mechanism for handling edge cases where fully autonomous policies fail, rather than serving as the default primary control mode. Furthermore, emerging brain-computer interfaces offer a potential pathway to bypass traditional motor bottlenecks, directly translating neural signals into robot commands to significantly expand the information throughput available for high-level supervision.

3.5 Discussion

Convergence toward hybrid architectures reflects a pragmatic division of labor where model-based layers enforce hard safety constraints while learning components address unmodeled dynamics and perceptual ambiguity [120, 123, 152, 162, 192]. Domain suitability dictates the specific coupling strategy: high-frequency dynamic tasks demand tight integration of learned residuals into convex optimization to handle hysteresis and variable inertia [7, 23, 133], whereas semantic manipulation benefits from hierarchical decoupling where foundation models generate latent action primitives for downstream physics-aware trackers [61, 95, 102, 132, 180, 182, 186, 195]. This evolution extends to human-in-the-loop frameworks, which increasingly treat motion retargeting as a distributional prior mapping problem rather than strict trajectory tracking, utilizing neural networks to bridge embodiment gaps without non-convex optimization [64, 201]. Such approaches position learning as the engine for adaptation and perception, while optimization remains the critical mechanism for ensuring constraint satisfaction in safety-critical deployments.

4 Challenges and Prospects

This chapter systematically examines the barriers preventing the broad adoption of whole-body control systems and proposes a strategic path for future advancement. Sec. 4.1 categorizes the fundamental limitations into six macro-dimensions, covering critical issues in dynamics modeling, computational efficiency, safety assurance, and generalization capabilities. Building on this analysis, Sec. 4.2 outlines a concrete research roadmap for the next decade, highlighting emerging technologies and necessary methodological shifts to overcome these bottlenecks. Together, these subsections provide a structured framework for understanding current deficits and directing subsequent innovation in the field.

4.1 Challenges

Whole-body control performance is currently constrained by a complex interplay of physical, computational, and safety-related barriers that prevent widespread deployment in unstructured settings. Fundamental discrepancies between simulated training environments and real-world hardware dynamics often degrade policy effectiveness upon transfer, as detailed in [Sec. 4.1.1](#). These physical challenges are compounded by strict real-time requirements, where the computational latency of high-dimensional optimization limits control loop frequencies ([Sec. 4.1.2](#)), while the absence of formal safety guarantees raises concerns regarding constraint violations under disturbance ([Sec. 4.1.3](#)). Furthermore, current methods struggle to generalize across different robot morphologies without extensive retraining ([Sec. 4.1.4](#)) and face significant difficulties in robustly coupling noisy perception data with low-level actuation ([Sec. 4.1.5](#)). Finally, ensuring safe and ergonomic physical interaction with humans remains a critical hurdle for deploying these systems in shared workspaces ([Sec. 4.1.6](#)).

4.1.1 Sim-to-Real Dynamics Discrepancy

The fidelity of learned policies degrades rapidly when deployed on physical hardware due to unmodeled non-linearities that simulation engines fail to capture. While synthetic environments assume idealized rigid bodies and perfect actuation, real-world systems exhibit complex compliance, latency, and friction profiles that create a substantial dynamics gap. These discrepancies invalidate the state estimates and control commands derived from sim-trained models, leading to instability even in tasks where the policy appeared converged.

1) Structural Compliance and Estimation Drift.

Standard rigid-body assumptions ignore elastic deformations in load-bearing links, causing significant state estimation drift that standard observers cannot correct [\[40, 59\]](#). This unmodeled flexibility violates binary contact assumptions, rendering terrain impedance estimation and gain-scheduled disturbance observers ineffective during dynamic interactions.

2) Actuator and Friction Model Mismatch.

Discrepancies in non-linear joint dynamics and contact friction between simulation engines like Isaac Gym or MuJoCo and physical hardware degrade tracking performance despite domain randomization efforts [\[36, 57, 103\]](#). Techniques such as actuator-net modeling or delta action residual learning often fail to fully compensate

for these differences, resulting in oscillatory behaviors or foot drift during zero-shot transfer.

3) Fragility of Adversarial Training Strategies.

Current adversarial training and domain randomization approaches frequently miss sparse, critical dynamic vulnerabilities, leading to policies that are either overly conservative or prone to catastrophic failure on complex terrain [\[139, 162, 199\]](#). Methods relying on privileged model-based guidance or visco-elastic environment modeling struggle to generalize against rotational spring singularities or non-zero-sum game perturbations that occur in reality.

4) Destabilization by Unmodeled High-Frequency Modes.

Simplified reduced-order dynamics often neglect high-frequency structural modes, such as foot compliance and link vibrations, which critically destabilize contact-rich locomotion policies [\[108\]](#). The absence of these fast dynamics in the training distribution causes controllers to excite resonant frequencies during impact, leading to immediate loss of stability rather than graceful degradation.

4.1.2 Computational Latency & Real-Time Constraints

Achieving control frequencies sufficient for dynamic stability often conflicts with the computational cost of resolving high-dimensional non-convex optimization problems. The inherent complexity of whole-body dynamics forces a compromise where either model fidelity is reduced or update rates drop below the threshold required for reactive balance.

1) Non-Convex Contact Complexity.

Incorporating accurate contact dynamics and complementarity constraints into nonlinear optimal control formulations creates prohibitive computational loads that restrict solution times to seconds or minutes [\[22, 33, 86\]](#). Solvers relying on direct multiple shooting or relaxed contact models frequently converge to local minima, rendering them unsuitable for online execution where rapid re-planning is essential.

2) Cubic Scaling in Centralized Solvers.

Centralized quadratic programming approaches exhibit cubic complexity relative to system degrees of freedom, establishing a hard ceiling on control frequency for robots with complex kinematic chains [\[4, 25, 165\]](#). The overhead from polyhedral approximations and actuator torque limits often triggers computation spikes during state transitions, while distributed consensus mecha-

nisms introduce additional latency that undermines real-time performance.

- 3) **Accuracy-Speed Trade-offs in Sampling.** Although sampling-based and learning-augmented methods sacrifice solution accuracy for speed, they still struggle to meet strict deadlines when managing high-dimensional action spaces or contact-rich scenarios [3, 66, 75, 104]. Issues such as low-accuracy algorithm truncation, inference latency in residual networks, and trajectory completion overhead in model predictive control often result in myopic planning horizons that fail to capture long-term dynamic requirements.
- 4) **Fragility of Latency Mitigation.** Strategies employing warm-starts or simplified kinodynamic models reduce average latency but become fragile during discontinuous contact events or aggressive maneuvers where prediction quality degrades [10, 77, 141]. Neural network warm-starts and zero-moment-point based models may fail to track velocity references during contact transitions, leading to torque controller failures and amplified sensor drift.

4.1.3 Safety Guarantees & Constraint Violation

Formal safety certification in whole-body control remains elusive because learning-based policies and optimization filters often lack rigorous bounds under arbitrary runtime conditions. While agile tracking is achievable, the absence of guaranteed constraint satisfaction exposes systems to catastrophic failures when facing unmodeled disturbances or distribution shifts.

- 1) **Inherent Guarantee Deficits in Learning.** Pre-trained reinforcement learning policies frequently fail to enforce arbitrary runtime constraints, leading to self-collision or posture remapping failures during adversarial attacks or actuator faults [113, 199, 205]. These methods exhibit high constraint violation rates when the operational environment deviates from training distributions, as they lack mechanisms to formally bound error propagation in hidden fault states.
- 2) **Horizon and Geometry Limitations in Optimization.** Safety filters relying on soft constraints or single-step barrier functions struggle to resolve non-convex geometric configurations and long-horizon discrete mode changes [26, 74, 152]. The infeasibility of relative-degree-two control barrier functions often forces reliance on reactive swing-leg reflexes, which cannot guarantee collision avoidance with non-convex prim-

itives or secure footholds on discontinuous terrain.

- 3) **Conservatism in Formal Stability Certificates.** Lyapunov and passivity analyses typically yield precision bounds that are excessively conservative or computationally intractable to determine a-priori under unknown input disturbances [82, 83, 157]. The ultimate error bounds depend heavily on adaptive gain tuning, rendering singular configuration instability difficult to predict without precise knowledge of the supremum of external perturbations.
- 4) **Fragility of Sensor-Driven Enforcement.** Constraint enforcement via tactile feedback is compromised by nonlinear sensor characteristics and significant errors in contact region estimation [83, 116]. Latency bottlenecks in updating dynamic environment maps, combined with servomotor backlash, introduce jitter that undermines the reliability of safety checks derived from LiDAR or proprioceptive data.

4.1.4 Morphological Generalization & Cross-Embodiment

Transferring control policies across distinct robotic morphologies remains a fundamental bottleneck because current architectures fail to disentangle task semantics from the specific kinematic and dynamic constraints of the training embodiment. The reliance on fixed observation and action spaces forces policies to overfit to the structural idiosyncrasies of a single robot, rendering them brittle when deployed on hardware with even minor topological variations. This fragility necessitates extensive retraining for every new platform, severely limiting the scalability of learning-based whole-body control.

- 1) **Kinematic Topology Mismatch.** Direct policy reuse collapses when agents encounter heterogeneous degrees of freedom or joint configurations that differ from the training distribution. Approaches attempting to align these structures via graph-based encoding or zero-padding often fail to capture the nuanced geometric dependencies required for closed-chain mechanisms or supernumerary limbs [126, 181, 185, 200]. The resulting representation gaps prevent the controller from generalizing motion primitives across varying skeletal layouts.
- 2) **Dynamic Parameter Sensitivity.** Policies trained on specific inertial profiles exhibit severe performance degradation when faced with unmodeled mass distributions or actuator torque

limits. Randomization techniques frequently overlook low-frequency variations in distal inertia or axial rotation dynamics, leading to instability when the physical robot deviates from the simulated domain [131, 140, 181, 185, 203]. Consequently, controllers lack the adaptability to handle payload changes or hardware wear without explicit fine-tuning.

- 3) **Fragile Intent Decoupling.** Separating high-level task goals from low-level coordination proves difficult due to residual coupling in contact management and gait selection strategies. Methods relying on embodiment-agnostic visual tokens or task-frame interfaces often suffer from drift in relative pose tracking or errors in command-space dimensionality [52, 118, 161, 179, 188, 193]. These abstraction failures cause the system to misinterpret whole-body constraints, resulting in inefficient or unsafe movements on unseen platforms.

4.1.5 Perception-Action Coupling in Unstructured Environments

Integrating noisy, high-dimensional sensory streams into low-level control loops creates a fragile coupling where perception artifacts directly destabilize whole-body dynamics. The latency inherent in processing visual data often exceeds the bandwidth required for reactive balance, while hierarchical decoupling allows errors to compound before reaching the actuator layer. Furthermore, limited sensor coverage and the sensitivity of unified policies to observation noise prevent reliable operation in cluttered, unstructured settings.

- 1) **Visual Latency and State Uncertainty.** Frame-based cameras introduce motion blur and depth estimation errors that corrupt state estimators, causing Extended Kalman Filters to drift during rapid maneuvers [101, 138]. This update rate mismatch violates rigid contact assumptions, leading to instability when the controller acts on outdated geometric information [40].
- 2) **Hierarchical Error Propagation.** Decoupling high-level planning from low-level execution creates a distillation gap where latent space inaccuracies manifest as dynamic failures [55]. Errors in termination probability prediction or foothold reachability accumulate across layers, resulting in catastrophic loss of balance during unstructured interaction [31, 202].
- 3) **Sensory Blind Spots and Global Awareness.** Localized sensing modalities fail to capture

global situational context, leaving non-sensorized links vulnerable to collisions due to self-occlusion [13]. These blind spots compromise multi-contact stability verification, often inducing oscillatory base movements when contact polygon estimates become unreliable [79, 116].

- 4) **Policy Sensitivity to Observation Noise.** Unified policies attempting to fuse high-dimensional exteroceptive data suffer from credit assignment conflicts that degrade performance in complex loco-manipulation tasks [61]. Advantage mixing becomes unstable under semantic reasoning latency, preventing effective suppression of grasp slips despite extensive domain randomization [51].

4.1.6 Human-Robot Interaction Safety & Ergonomics

Physical collaboration between humans and robots introduces unique failure modes where control stability and operator well-being are directly compromised by unmodeled dynamic couplings and interface fidelity limits. While theoretical frameworks often assume idealized contact conditions, real-world deployment reveals critical fragilities in how systems estimate external forces and map human posture to robotic objectives during transient interactions.

- 1) **Fragile Disturbance Estimation.** Maintaining safe impedance during dynamic contact mode transitions or floating-base motions remains precarious due to bandwidth mismatches between proprioceptive force estimation and actual physical interaction speeds [16, 90]. Configuration-invariant normalization techniques and Kalman covariance inflation often fail to suppress vibrations when sim-to-real force overshoot occurs, leaving the system vulnerable to instability during sudden load changes [128, 159].
- 2) **Static Ergonomic Modeling.** Current optimization frameworks frequently rely on discrete REBA scoring and constant trunk assumptions that neglect coupled rigid-body dynamics, resulting in suboptimal kinetic energy minimization during collaborative tasks [89, 135]. These simplified kinematic models cannot capture dynamic load distribution, leading to generic supernumerary limb weight penalties that misrepresent the true physiological cost to the human operator [133, 154].
- 3) **Interface Transparency Deficits.** Teleoperation architectures suffer from high cognitive load induced by unilateral visual feedback loss and

tactile sensing calibration drift, which sever the intuitive link between command and physical outcome [8, 112]. The burden of customizing deadband zones combined with inertial transparency issues from mobile base transmission further degrades performance, while VLM inference latency disrupts the temporal synchrony required for safe physical assistance [41, 145, 171].

4.2 Prospects

The future trajectory of whole-body control hinges on integrating large-scale foundation models to achieve semantic generalization across diverse tasks and embodiments, as detailed in Sec. 4.2.1. To address latency constraints in dynamic environments, upcoming research will explore the fusion of neuromorphic sensing with rapid reactive loops (Sec. 4.2.2), while simultaneously developing self-supervised mechanisms for continuous dynamics adaptation during operation (Sec. 4.2.3). The scope of control frameworks is expected to expand from single agents to collaborative multi-robot systems capable of shared physical manipulation (Sec. 4.2.4), necessitating new strategies that prioritize energy efficiency and passive compliance (Sec. 4.2.5). Finally, the field requires unified benchmarking protocols to rigorously evaluate stability and performance across these emerging platforms (Sec. 4.2.6).

4.2.1 Foundation Models for Embodied Control

Scalable motion tracking objectives are poised to replace manual reward engineering by providing dense, universal supervision signals for whole-body control [107, 190]. This shift eliminates heuristic reward shaping, establishing trajectory reproduction as a foundational task that follows observed scaling laws for behavioral diversity. Unified token spaces and global frame representations will further enable policies to generalize across heterogeneous embodiments and unseen motion distributions without retraining [107, 190]. Techniques such as vector quantization for cross-embodiment alignment and latent space structuring via normalization layers facilitate zero-shot transfer capabilities. Furthermore, learning from imperfect video-derived references through residual action grounding will allow robots to acquire complex interaction skills without precise motion capture data [174]. By utilizing residual action spaces and contact-promoting interaction rewards, these systems achieve robustness to reference noise while enabling monocular video imitation.

Structured latent spaces must directly align vision-language semantics with high-degree-of-freedom kine-

matic trajectories to replace hand-crafted action vocabularies [14, 180]. Architectures will learn modality-invariant features that eliminate rigid end-effector pose restrictions through adversarial objectives specifically designed to enforce consistency with learned motion priors. Cross-embodiment generalization arises from grounding policies in optimal control-derived priors, where iterative visual feedback loops correct morphological discrepancies between human demonstrations and robot hardware [14, 67]. This process integrates trajectory generation mechanisms with vision-language model-based refinement to address embodiment-specific errors without relying on specific parametric human models. Hierarchical designs will decouple high-level semantic reasoning from low-level dynamics execution, enabling zero-shot sim-to-real transfer while maintaining stability through policies distilled from synthetic kinematic demonstrations [51, 180]. The incorporation of whole-body tactile sensing with ego-centric visual observations resolves contact ambiguities, providing the dense multimodal grounding necessary for complex dynamic maneuvers [14, 51].

Resolving the frequency mismatch between autoregressive generative planning and high-frequency control loops requires asynchronous execution frameworks that utilize trajectory chunking and temporal ensembling [20]. Such architectures decouple latency from jitter elimination, enabling stable replanning without disrupting low-level actuation commands. To ensure robust closed-loop recovery from physical disturbances, training protocols must expose policies to self-generated noisy prefixes through scheduled sampling curricula [20, 156]. This approach mitigates cumulative exposure bias by forcing the policy to learn error-recovery strategies conditioned on imperfect states rather than idealized trajectories. Furthermore, structuring latent spaces with forward-backward representations and stability priors derived from human motion supports generalization across unseen tasks by embedding stability constraints directly into the latent geometry [156, 198].

4.2.2 Neuromorphic & Event-Based Sensing Integration

Transitioning from frame-based visual processing to asynchronous event-driven streams offers a direct pathway to eliminate motion blur and drastically reduce perception latency during high-speed perturbations. By encoding temporal contrast through sparse spike generation, these sensors achieve microsecond-level response times and expanded dynamic range, enabling control loops to react before traditional cameras can even capture a coherent image [13, 39].

This sensory shift allows for the fusion of neuromorphic vestibular and proprioceptive signals, creating multi-modal weighting strategies essential for training anticipatory postural adjustments under rapid disturbance conditions. Furthermore, integrating such sparse proximity data with full-body geometric models facilitates the synthesis of virtual control points on non-sensorized links, permitting whole-body collision avoidance without requiring dense physical sensor coverage [13]. The convergence of these mechanisms suggests a future where reactive balance relies on continuous, low-latency spike trains rather than discrete frame updates to maintain stability against unpredictable kinetic stepping forces.

Architectural compatibility with control policies necessitates a hierarchical decoupling where sparse event streams drive high-frequency tracking while semantic reasoning operates on slower timescales [132, 180]. This dual-process structure aligns multimodal sensory data with kinematic trajectories through asynchronous processing pipelines, preventing signal degradation during rapid policy execution without relying on complex generative latent spaces. Bio-inspired frameworks further utilize spike-based integration mechanisms conditioned on these temporal vectors to synthesize smooth motion segments that remain compatible with microsecond-level sensor updates [132]. Ensuring dynamic stability requires propagating physical tracking errors directly through these asynchronous pathways, thereby closing the sim-to-real gap and enabling latency-aware optimization for high-speed feedback loops.

Prioritizing sparse event-driven data transmission will become essential for future onboard deployment to minimize memory bandwidth consumption. By encoding sensory information asynchronously, embedded compute units can avoid the thermal loads associated with continuous frame-based processing during extended operation. This shift toward dynamic range compression and sparse data representation directly addresses the power constraints limiting current mobile platforms. Consequently, control systems will sustain higher update frequencies without exceeding the thermal dissipation limits of compact robotic chassis.

4.2.3 Self-Supervised Dynamics Adaptation

Integrating lightweight neural residuals directly into analytical dynamics models offers a pathway to correct unmodeled physical phenomena, such as aerodynamic spin or material hysteresis, in real time [23, 120]. This hybrid architecture enables the controller to learn specific error patterns while preserving

the structural guarantees of the underlying physics-based model. To handle abrupt changes in payload or environmental interaction forces, these systems must achieve rapid convergence of dynamic estimates within single control episodes [23, 120]. Such few-shot system identification capabilities allow robots to adapt to new conditions without requiring extensive offline retraining or prolonged data collection phases. The resulting framework supports sub-second error convergence through piecewise key feature representation and continuous constraint enforcement.

Embedding predictive world models into whole-body controllers offers a pathway to anticipate long-term kinematic drift and structural wear by synthesizing sparse task-space references with dense internal state projections [104, 193]. Such architectures utilize trajectory completion mechanisms and sliding-window global rewards to maintain implicit spatiotemporal alignment during extended policy-in-the-loop rollouts, effectively reducing the command space required for stable planning. By modifying only high-level cost functions or object models within the predictive horizon, these systems can achieve zero-shot adaptation to changing payload dynamics and varying object properties without retraining low-level policies [193]. This hierarchical integration of model predictive control and reinforcement learning facilitates efficient sample-based planning that accommodates test-time generalization through direct object model substitution. The resulting framework enables robots to autonomously compensate for physical degradation while preserving task performance across diverse operational conditions.

4.2.4 Collaborative Multi-Robot Whole-Body Systems

Transitioning from treating inter-agent interactions as external disturbances to modeling them as internal coupling terms within a unified floating-base dynamic equation defines the next theoretical frontier for multi-robot systems. Future frameworks must embed coupled rigid-body dynamics directly into the control formulation, enabling centralized momentum regulation and bidirectional force optimization that treats physical contact as an intrinsic system property rather than a perturbation [134, 167]. Achieving global stability in such tightly connected teams requires extending hierarchical task prioritization to resolve shared contact constraints and individual postural ergonomics simultaneously within a single optimization layer. This architectural unification allows the prioritized control structure to enforce friction limits and minimize torque norms while maintaining mathematical separation for conflicting objectives

[134, 167]. The resulting solution space separates distinctly into motion subspaces for trajectory tracking and constraint subspaces dedicated to maintaining force-closed grasps across underactuated legged teams. Such separation supports impedance strategies that explicitly account for underactuation constraints and projection conditions necessary for stable cooperative manipulation [167].

Replacing rigid centralized planners requires hybrid frameworks that decouple high-level coordination invariant to spatial transformations from reactive low-level execution to ensure generalization across varying team configurations [93, 167]. Such architectures utilize canonicalized task space representations to manage underactuated dynamic projections without relying on global state synchronization. Communication efficiency further demands integrating proactive kinematic planning with reactive compliance layers that locally dissipate internal forces while prioritizing the primary task objective [93, 167]. This approach regulates contact wrench distribution while maintaining stability despite latency-induced trajectory deviations. Robust synchronization in floating-base teams ultimately necessitates embedding constraint enforcement mechanisms directly into local optimization routines to maintain closed-chain integrity [93, 167]. These routines must simultaneously satisfy physical interaction limits and actuator capacity constraints to compensate for kinematic drift in real time.

4.2.5 Energy-Efficient & Compliant Actuation Control

Morphological symmetry offers a pathway to uniformize center-of-mass acceleration feasibility, thereby reducing the computational burden associated with direction-specific stability margins [100]. By adopting radially oriented linear actuators within a dodecahedral distribution, future architectures can achieve dynamic isotropy that ensures omnidirectional stability with uniform control effort. This spatial redundancy allows systems to saturate energy efficiency gains by balancing control loads across all axes without requiring explicit trajectory re-planning [100]. Exploiting such passive compliance through symmetric actuation topologies inherently enhances disturbance resilience while minimizing active correction costs.

Explicitly optimizing transmission topology reconfiguration alongside joint trajectories represents a critical frontier for minimizing distal inertia and maximizing energy efficiency during operational mode switches. Future frameworks will likely treat base-mounted actuation and belt transmission dynamics as decision

variables, enabling active steering mechanisms to outperform passive feet in specific locomotion regimes [140]. This approach necessitates control architectures that simultaneously solve for mechanical state and structural arrangement, effectively reducing the energetic cost of accelerating limb segments. By strictly coupling these hardware-aware planning objectives with compliant actuator dynamics, systems can achieve optimal energy distribution without relying on external policy corrections. The resulting synthesis ensures that efficiency gains derive directly from the co-optimization of physical configuration and control effort rather than compensatory learning strategies.

4.2.6 Standardized Benchmarking & Evaluation Protocols

Decoupling high-level policy generalization from low-level tracker dynamics remains essential to isolate execution failures during leg-critical interactions. Future protocols should enforce a hierarchical architecture with a canonical action interface, enabling the detection of cross-embodiment transfer asymmetries under perturbation-conditioned scenarios [169]. Standardized task suites must categorize manipulation modalities to quantify stability trade-offs between locomotion autonomy and payload capacity without prescribing rigid taxonomic structures. This approach requires embedding evaluation criteria that penalize center-of-mass shifts while respecting physical contact constraints and whole-body grasping limits [49]. Evaluation frameworks need controlled stress tests distinguishing semantic understanding deficits from physical feasibility violations during dynamic contact switches. Such metrics would combine visual distractor sensitivity with force-closure validation to produce retention rates that reflect true system robustness [49, 169].

Transitioning from proprietary motion capture setups to accessible commercial gaming consoles offers a scalable pathway for quantifying whole-body tracking fidelity. Proposed frameworks suggest integrating console-based scoring systems as objective metrics, where hardware-in-the-loop benchmarking validates monocular pose reconstruction against kinematic re-targeting pipelines. Such approaches must evolve beyond simple geometric alignment to include dual-metric structures that simultaneously assess tracking accuracy and dynamic stability penalties under high-frequency control constraints. Future protocols should aim to establish correlations between score errors and task success rates by combining these console-based evaluations with dynamic stability assessments, thereby revealing specific vulnerabilities

in upper-body exploitation during complex maneuvers. Enforcing these combined measures will ensure smoothness penalties accurately reflect physical feasibility rather than mere visual similarity.

Decoupling algorithmic logic from hardware-specific interfaces through cloud-native architectures offers a viable path toward reproducible cross-embodiment validation. Future infrastructures must enforce unified state-space formulations and abstract hardware protocols to facilitate modular planner-controller swapping without re-engineering core policies [169]. Achieving this requires shared simulation backends and communication middleware that allow identical control policies to deploy across heterogeneous robotic morphologies using canonical action interfaces. Integration of established kinematics engines and physics solvers with standard networking stacks will enable controller state machine coordination across diverse platforms. Benchmark suites should further adopt hierarchical decomposition to isolate high-level policy prediction capabilities from low-level motion tracking adapters during transfer assessments. This structural separation allows researchers to pinpoint whether failures stem from cross-morphology asymmetry or inadequate perturbation handling [169].

5 Conclusion

Whole-body control is transitioning from a stability-centric engineering problem to a learning-driven autonomy challenge, where dynamic motion capabilities will increasingly function as standardized outputs derived from the foundation model analyses presented in Sections 3 and 4. Realizing this vision requires hybrid architectures that combine the data efficiency of model-based optimization with the behavioral adaptability of learned policies. Widespread adoption depends on resolving critical gaps in safety certification, cross-embodiment transfer, and standardized evaluation. Beyond these technical hurdles, deploying such dynamic systems in human environments necessitates a rigorous examination of ethical implications to ensure autonomous behaviors align with societal safety norms and accountability standards. Addressing these intertwined technical and ethical dimensions is essential to accelerate the reliable integration of advanced robots into real-world settings.

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